

A2 —Domain Model and Testable Specification

Course **Intelligent Software Engineering**, UCAS, 2026 autumn term · Platform <https://le.arn.spaiq.ai> Author: 汤力为 (Liwei Tang), student ID 2026E8001082052 Repository: <https://github.com/Tom1246/ise-course> · tag a2-v1 · report generated 2026-10-04

This report is the reading copy of A2. Its sources are the files below, kept next to it as the authoritative versions; the executable parts are a check script, a notebook whose outputs are stored in it, and diagrams whose Graphviz sources are committed alongside the rendered images.

Part	Source file	Executable / regenerable with
Domain glossary and context map	domain-glossary.md	bash figs/render.sh (context map)
Data dictionary	data-dictionary.md	measured from the frozen inputs
Invariants and checks	invariants.md	python3 scripts/a2_check_invariants.py
One material rule, three ways	conflict-decision-table.md	check script, section 1
Data source, authority, quality, retention, deletion, privacy	data-governance.md	—
Model walkthrough record	model-walkthrough.md	notebooks/a2-domain-model-walkthrough.ipynb
Raw output of the checks	—	data/derived/a2/invariant-checks.md

A2 —Domain glossary and context map

The ubiquitous language for the course-planning domain. Terms are given Chinese-first (the data and the rules are Chinese) with the English term that code and documents use. Where a term has a trap, the trap is stated —half of this domain's bugs come from words, not from algorithms.

1. Bounded context

Semester course planning —from the university says what will run to the student has a timetable and an order to register in. Everything in this repository lives inside this boundary.

- **Upstream, outside:** the university's course-offering system (publishing the plan and the listing) and the programme requirements document. We consume them; we cannot

change them.

- **Downstream, outside this context:** the registration transaction itself. Registration is not automated here —out of scope by policy (see `data-governance.md` and A1's misuse scenario).
- **The unit of work** is one student, one semester, one plan. Not a cohort, not a department.

2. Entities —identity that persists across time

课程

- English: Course
- Identity: 课程编码 (18 characters)
- What it is: a course as a catalogue entry
- Traps found in the real data: the code is not opaque: position 13 = 培养层次, 14 = 课程属性, **18 = 校区 (Y = 玉泉路)**

班次

- English: Section
- Identity: 课程编码 + suffix (**-01**, **-08**, **-W003**)
- What it is: one teaching instance with its own slot, room, capacity
- Traps found in the real data: one course normally has several sections with different slots —“ the course clashes” is almost always false; sections clash

学期

- English: Term
- Identity: e.g. 2026-2027 秋季
- What it is: the teaching period a section belongs to
- Traps found in the real data: the workbook **does** carry it, per row (秋季 400 / 春季 332 in the committed import) —A1 used exactly that column to show the three required courses are autumn-only; the public listing has no per-row term (the term is implied by the crawl)

培养方案

- English: ProgrammeRequirement
- Identity: 校发培养字 (2025) 92 号 + programme code
- What it is: the credit floors and degree-course rules a plan must satisfy
- Traps found in the real data: states floors only; no ceiling (see `invariants.md`, l2)

选课方案

- English: Plan
- Identity: the student's own
- What it is: the chosen sections for one term, plus an order to register in
- Traps found in the real data: **personal data** —kept out of this repository (aggregates only)

选课记录

- English: Enrollment
- Identity: student × section
- What it is: the state of one student's registration for one section
- Traps found in the real data: the privacy-sensitive entity: enrolment lists identify people

3. Value objects —no identity, equal by value

Chinese	English	Value	Notes
周次区间	WeekRange	(start, end), e.g. (2, 5)	the field is an interval set , not a number: 第 2–5, 7–17 周 is [(2, 5), (7, 17)]
节次区间	PeriodRange	(first, last), e.g. (10, 12)	节 are 1..12 within a day
星期节次	DayPeriod	(weekday, PeriodRange)	the listing packs several into one cell: 周三 (5–7); 周六 (5–7)
时段	Slot	DayPeriod + WeekRange set	the unit the conflict rule compares
学分	Credit	number	used by I2
容量 / 已选	Capacity / Enrolled	non-negative integer or unknown	0 is a sentinel for "not set" —40 rows in the snapshot have 0 capacity with people enrolled
教室	Room	string	informational; not part of any invariant
课程属性	CourseAttribute	学位课 / 选修 / 必修	the degree-course floors are expressed in this vocabulary
校区	Campus	Y = 玉泉路, H = 雁栖湖, Z = 中关村	centralised-teaching students may not cross campuses (school notice, quoted in A1)
冲突	Conflict	boolean derived from two Slots	derived, never stored —see the decision table

4. Domain events —what happens, in the order it happens

Event	Trigger	Meaning for the model
CoursePlanPublished	the university issues the workbook (2026-08-28)	sections get a planned existence; no weekday/weeks yet
CatalogueSnapshotTaken	we crawl the listing (2026-09-27)	a frozen view; every later question is answered against it

Event	Trigger	Meaning for the model
RegistrationWindowOpens	09-02 public electives, 09-03 professional, 09-04 12:30 for first-year master's , 09-18 12:30 close	the order to register in becomes the scarce resource
SectionFull	enrolled = capacity	the section is effectively unavailable; the plan must change
SectionOverCapacity	enrolled > capacity > 0	a state the system allows; must be surfaced, never read as "a seat exists"
SectionSelected / SectionDropped / SectionSwitched	student action	the real case in A1 : switching a full section re-arranged the whole timetable
PlanRevised	student action	A1's two saved versions are two instances of this event (22 removed / 45 added cells)
SnapshotStaled	time passes	the catalogue no longer matches the live system: the plan is a hypothesis, not a fact

5. Context map

See `figs/context-map.dot` (source) and `figs/context-map.png` (rendered; regenerate with `bash doc/assignments/a2/figs/render.sh`). Reading it: two upstream contexts (offering, requirements) feed the planning context; the planning context emits a timetable plus a registration order; the registration context is downstream and deliberately outside the automated boundary.

6. Words we refuse to use loosely

- " 课 " **alone** —always say 课程 (course) or 班次 (section). A1's own first draft confused the two, and the confusion is what makes " the course is full " sound true when only one section is.
- " 优先 " / **priority** —two different orders exist and they are not the same: the **plan order** (which course matters more when something must be dropped) and the **grab order** (which section to register first under first-come-first-served). Documents say which one is meant.
- " 抢课 " —the act of competing for seats. This project outputs an order, never an actuator.
- " 周次 " —never a single number; always an interval set. 第 6 周 is the degenerate set $[(6,6)]$.

- ” 满” (full) —a state of a section, not a violation of an invariant (see `invariants.md`, l3).

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A2 —Data dictionary (S1 / S2 / S5, field by field)

A field-level reference for the three frozen inputs this project consumes, written to be looked up, not read end-to-end. Field names are given Chinese-first (the data is Chinese) with the English name code and documents use; the semantics are constrained to `doc/assignments/a2/domain-glossary.md` (entities: 课程/班次/学期/培养方案; value objects: 周次区间/节次区间/星期节次/时段/学分/容量/教室/课程属性/校区).

- **Every number below was measured in this session** with inline python3 on the committed files (no number is copied from another document). The exact snippet is cited next to each figure under ” 实测出典” / in §5. No script file was added (the A2 brief allows only this one new file), so the measurements are reproducible from the quoted one-liners and the frozen hashes in §0.2.
- Companion raw output: `data/derived/a2/invariant-checks.md` (§0 ” Parsing quality”); where I re-checked its numbers, §5 records both and flags any difference.

0. Scope, sources and how to read this

0.1 Sources and the naming caveat (read this first)

label	what it is	committed file(s)	rows
S1	开课计划表—official course-planning workbook (计划态)	<code>data/s1-plan-workbook/sections-snapshot.json</code>	732
S2	教务公开课表—official public listing, <code>jwbaucas.ac.cn</code> (当前态, crawled 2026-09-27)	<code>data/s2-public-listing/89576-campus20.json</code> (+ <code>89577-campus20.json</code> , history/)	341 (autumn)
S5	培养方案—校发培养字〔2025〕92号 credit floors	<code>data/s5-programme/degree-requirements.json</code>	1 object
(derived)	课程库快照—the 347-course 玉泉路 dataset that A2's conflict check actually runs on	<code>data/derived/course-library-snapshot.json</code>	347

Naming caveat (measured, not a typo). This repository's convention (README.md §” Where the data comes from” , A1 brief) is: **S1 = 开课计划表**, **S2 = 教务公开课表 jwba**. The

347-row `course-library-snapshot.json` is neither: it is a **snapshot of course data crawled from the academic system**, archived through the course-planning site's public JSON (`default-courses.json`; the URL, size and hash of the archived copy are recorded in `data/derived/live-listing-provenance.md`, and the same data is what `doc/data-authority.md` calls source (表情)). It is a **derived** dataset —a second pipeline over the same academic-system facts —so it is used for cross-checking and as the carrier of the schedule fields the A2 checks consume, **never as an authority**. Because it is derived, it is not expected to be identical to S2; where the two disagree on the 251 shared base codes (weekday/period 8 cases, week sets 9 cases $\approx 3\%$) this document records the disagreement rather than smoothing it (§ 5.6, § 6 T8/T10).

0.2 Frozen-input hashes (recomputed this session)

file	sha256 (recomputed)	committed .sha256
<code>data/derived/course-library-snapshot.json</code>	3724ade7ff05dcd3465c017f965f6c1fed0a426	5f6c1fed0a426c6fbcaafebf976d2680031
<code>data/s1-plan-workbook/sections-snapshot.json</code>	ffcf1ae023e59fe8a09d83584f77dab29	82cce8887f51e7db140f98
<code>data/s2-public-listing/89576-campus20.json</code>	6510aa2e4127a096208de7d02916c6bc5	3f66f050d68470cff2d833
<code>data/s5-programme/degree-requirements.json</code>	36f99278312e4954129c9bd1575a794859d209840268bfc	512a061a (表情) (source PDF hash recorded in <code>data/s5-programme/provenance.md</code>)

0.3 Envelopes (the files are not bare arrays)

file	top-level shape	how to read it
<code>course-library-snapshot.json</code>	<code>{"campus_rule": "course code position 18 == 'Y'", "courses": [... 347 ...]}</code>	rows are objects; one row = one 班次
<code>sections-snapshot.json</code>	<code>{"campus_filter": " 玉泉路", "courses": [... 732 ...]}</code>	rows are objects; one row = one workbook record

file	top-level shape	how to read it
89576-campus20.json	<pre>{ "campus_code": 20, "term_id": 89576, "crawled_at": "2026-09-27 16:31:49", "source": "https://jwbaucas.ac.cn/sc/public/coursePublic", "rows": [... 341 ...] }</pre>	rows are objects; one row = one 班次
degree-requirements.json	<pre>{ "_source": "degree", "programme": "degree_track", "semester_min_credits": 16, "categories": "category", "known_modelling_gaps": 0 }</pre>	rules, not rows

1. S1 — 开课计划表 (course-planning workbook)

data/s1-plan-workbook/sections-snapshot.json — 732 records, 414 distinct 课程编码 (413 after the one section suffix is stripped), campus filter 玉泉路. Source workbooks: 《…开课计划表 0828.xlsx》441 + 《…核心课和专业课列表 0828.xlsx》291.

课程编码 / code

- 所属实体: 课程 (Course) + 班次 (Section)
- 类型与单位: string; 18 字符为基码, 可带 -xxx 班次后缀
- 取值 / 示例: 180080025200M1002Y (18 位)、180089050200MB001Y-201 (22 位, 唯一带后缀的一条)
- 来源: S1
- 质量与陷阱: 不是数字, 不能当整数排序。第 18 位恒为 Y — 实测 732/732 的第 18 位都是 Y (玉泉路)。前缀 18 位是课程身份, - 之后是班次身份。

课程名称 / name

- 所属实体: 课程
- 类型与单位: string
- 取值 / 示例: 抽样调查
- 来源: S1
- 质量与陷阱: 非唯一; 不同班次同名, 且同名课程在不同编码下存在。

学分 / credit

- 所属实体: 学分 (Credit)
- 类型与单位: number (本文件为 JSON number: 整型 582 条 / 浮点 150 条)
- 取值 / 示例: 2、0.5、2.5、3.5 (取值集合 {0.5, 1, 2, 2.5, 3, 3.5})

- 来源: S1
- 质量与陷阱: 与 S2 的字符串型 credit ("2.00") 类型不同, join 前须归一化; 是 I2 唯一计分来源。

开课院系 / department

- 所属实体: 班次
- 类型与单位: string
- 取值 / 示例: 数学科学学院; top: 国际学院 160、计算机科学与技术学院 87、电子电气与通信工程学院 87、外语系 73
- 来源: S1
- 质量与陷阱: 开课单位, 不是” 课程所属学科” ; S2 有独立的 department 语义相同但口径是” 开课单位” 。

开课学期 / semester

- 所属实体: 学期 (Term)
- 类型与单位: enum string
- 取值 / 示例: 秋季 (400 条)、春季 (332 条)
- 来源: S1
- 质量与陷阱: **实测 S1 里确实有该字段** (见 §6 陷阱 T7 的反例说明)。S2/快照都没有逐行学期; S2 的学期只能从 term_id 按文件推。

课程属性 / attribute

- 所属实体: 课程属性 (CourseAttribute)
- 类型与单位: enum string
- 取值 / 示例: 专业课 234、学科核心课 206、专业核心课 142、公共选修课 79、公共必修课 29、研讨课 25、实验课 16、实践课 1
- 来源: S1
- 质量与陷阱: 取值集合有 8 个, 但**快照/S2 出现 科学前沿讲座 而没有 实践课** (§5.7) → 两源属性词表不完全一致, 不能当作同一受控词表 join。

开课校区 / campus

- 所属实体: 校区 (Campus)
- 类型与单位: enum string
- 取值 / 示例: 玉泉路 (732/732)
- 来源: S1
- 质量与陷阱: 本文件已被 campus_filter= 玉泉路 预过滤 → **没有区分度**, 不能用来做跨校区判断; S2 的 campus 才是逐行真字段。

限选 (容量) / capacity

- 所属实体: 容量 (Capacity)
- 类型与单位: 非负整数字符串
- 取值 / 示例: **全部为空** (732/732 空串)
- 来源: S1

- 质量与陷阱: **S1 完全没有容量数据**。空串 $\neq 0 \neq$ 无限; 缺失必须显式建模 (domain-glossary: Capacity = non-negative integer **or unknown**)。

已选 / enrolled

- 所属实体: 已选 (Enrolled)
- 类型与单位: 非负整数字符串
- 取值 / 示例: **全部为空** (732/732 空串)
- 来源: S1
- 质量与陷阱: 同上; S1 无法支持 I3。

分段排课 / sessions

- 所属实体: 班次 $\rightarrow$ Slot[]
- 类型与单位: array of object, 长度恒为 1
- 取值 / 示例: [{"room":"","time":"","weeks":""}]
- 来源: S1
- 质量与陷阱: **实测 732/732 都是长度恰为 1 的空对象** (room 也全空) $\rightarrow$ 该字段是空壳, 不能据此认为“S1 有排课结构”。

星期节次 / time

- 所属实体: 星期节次 (DayPeriod)
- 类型与单位: string (应为 周一 (1-2); ...)
- 取值 / 示例: **全部为空**
- 来源: S1
- 质量与陷阱: 与 provenance 一致 (0/732 有值) $\rightarrow$ 单靠 S1 无法做冲突判断。

开课周 / weeks

- 所属实体: 周次区间集 (WeekRange set)
- 类型与单位: string (应为 第 2-5,7-17 周; ...)
- 取值 / 示例: **全部为空**
- 来源: S1
- 质量与陷阱: 同上。A1 报“0/732”, **本次复核仍为 0/732** (§ 5.4)。

考试方式 / exam

- 所属实体: 考试方式 (ExamMode)
- 类型与单位: string
- 取值 / 示例: **全部为空** (732/732)
- 来源: S1
- 质量与陷阱: S1 无考核方式; 快照有 (闭卷笔试 等 6 值)。

来源工作簿 / source

- 所属实体: 记录溯源元数据
- 类型与单位: string

- 取值 / 示例: 2026–2027 学年秋季和春季开课计划表 0828.xlsx (441)、2026–2027 学年研究生核心课和专业课列表…0828.xlsx (291)
- 来源: S1
- 质量与陷阱: **溯源自证**: 一条记录来自哪本工作簿, 可用来解释属性/字段覆盖差异; 不是课程属性。

S1 缺失字段 (三个来源都没有、或只有别处有): level (培养层次)、subject (所属学科/专业)、teacher (主讲教师)、hours (学时) 在本文件均不存在。

2. S2 —教务公开课表 (public listing, jwba)

2a. 文件级字段 (89576–campus20.json 的顶层)

校区码 / campus_code

- 所属实体: 抓取参数
- 类型与单位: integer
- 取值 / 示例: 20
- 来源: S2
- 质量与陷阱: 20= 玉泉路 (21= 中关村、22= 雁栖湖, 见 doc/data-authority.md)。整文件一个值 → 不是逐行字段。

学期号 / term_id

- 所属实体: 学期 (Term)
- 类型与单位: integer
- 取值 / 示例: 89576
- 来源: S2
- 质量与陷阱: 映射见 terms.json: 89576 = 2026—2027 学年 (秋) 第一学期, 89577 = (春) 第二学期。**学期只能由 term_id 反查, 行内无学期字段。**

抓取时间 / crawled_at

- 所属实体: 记录时点
- 类型与单位: datetime string (local)
- 取值 / 示例: 2026–09–27 16:31:49
- 来源: S2
- 质量与陷阱: 这是 SnapshotStaled 事件的证据; S1 的时点是文件名里的 0828 (2026-08-28)。

来源 URL / source

- 所属实体: 溯源元数据
- 类型与单位: string (URL)
- 取值 / 示例: https://jwbaucas.ac.cn/sc/public/coursePublic
- 来源: S2
- 质量与陷阱: 列表页地址; 详情页是 https://jwbaucas.ac.cn/sc/course/coursetime/{course_id} (脚本 scripts/fetch_official_db.py)。

行集 / rows

- 所属实体: 班次 (Section)[]
- 类型与单位: array of object, 长度 341
- 取值 / 示例: 见 § 2b
- 来源: S2
- 质量与陷阱: 89577-campus20.json 的 rows 长度 **实测为 0** (春季尚未发布) → 拿去 join 会静默得到空集。

抓取页哈希 / schedule_pages_sha256 (行内)

- 所属实体: 溯源元数据
- 类型与单位: string (hex)
- 取值 / 示例: 每行一个
- 来源: S2
- 质量与陷阱: 逐行记录该班次详情页快照哈希, 可复算; 证明这一行来自哪一次抓取。

字段集口径提醒: data/derived/live-listing-provenance.md 记录上游原始下载 (8,103,455 字节, 未入库) 的字段是 **19 个 SEP「学期课表」表头**; 而 **committed 的 89576-campus20.json 每行实测只有 15 个 key** (下表)。被裁掉的上游字段包括 培养层次/授课方式/考试方式/是否远程教学/所属学科·专业/助教——其中 level、exam、subject 改由 § 2c 的 347 快照承载。” **19 字段” 指上游原始下载, 不是这个 frozen 文件。**

2b. 行级字段 (rows[i], 实测 15 个 key, 341 行; 空值对数 = 该字段空串条数)

序号 / seq

- 所属实体: 抓取序号
- 类型与单位: string(数字)
- 取值 / 示例: "1"、"2" ...
- 来源: S2
- 质量与陷阱: 页内序号, 非班次身份; 不要当主键。

开课单位 / department

- 所属实体: 班次
- 类型与单位: string
- 取值 / 示例: 数学科学学院 (空 0/341)
- 来源: S2
- 质量与陷阱: 与 S1 department 语义可对齐。

课程编码 / code

- 所属实体: 课程 + 班次
- 类型与单位: string
- 取值 / 示例: 180080070100MX018Y、180089050200MB001Y-304 (空 0/341)
- 来源: S2
- 质量与陷阱: 同 S1: 第 18 位恒为 Y (实测 341/341)。

课程名称 / name

- 所属实体: 课程
- 类型与单位: string
- 取值 / 示例: 实用最优化算法 (空 0/341)
- 来源: S2
- 质量与陷阱: —

课程属性 / attribute

- 所属实体: 课程属性
- 类型与单位: enum string
- 取值 / 示例: 公共选修课 47、学科核心课 75、专业课 72、专业核心课 53、公共必修课 78、研讨课 8、实验课 7、科学前沿讲座 1 (空 0/341; 合计 341)
- 来源: S2
- 质量与陷阱: 与 S1 属性词表略有出入 (S2 有 科学前沿讲座, S1 有 实践课)。

学分 / credit

- 所属实体: 学分 (Credit)
- 类型与单位: **string**, 两位小数
- 取值 / 示例: "1.00"、"2.00"、"2.50"、"0.50"; 分布 2.00138/1.0078/3.0070/2.5034/0.5019/3.502 (空 0/341)
- 来源: S2
- 质量与陷阱: 字符串型! join 前必须 float(); 与 S1 的 number 型不同 (§1)。

课时 / hours

- 所属实体: 学时 (Hours)
- 类型与单位: string(整数)
- 取值 / 示例: "40"99、"60"57、"30"28、"128"22 (空 0/341)
- 来源: S2
- 质量与陷阱: **不是学分**; 与 credit 同源一个上游列” 课时·学分” 被拆开。S1/S5 无此字段。

限选 (容量) / capacity

- 所属实体: 容量 (Capacity)
- 类型与单位: 非负整数字符串 或 "/"
- 取值 / 示例: 数值 301 行 (4047、5847、3032...); "/" 40 行 (空 0/341)
- 来源: S2
- 质量与陷阱: **官方” 未设置” 哨兵是 "/"**, 不是 0 (§5.3)。"/" 绝不能当容量 0 读——否则 40 行会被误判成” 容量为零”。

已选 / enrolled

- 所属实体: 已选 (Enrolled)
- 类型与单位: 非负整数字符串
- 取值 / 示例: 075 行、111、149... (空 0/341; 无 "/")
- 来源: S2

- 质量与陷阱：值 0 是真实“没人选”，与 capacity 的哨兵语义**不同**；两列不要用同一套缺失规则。

分段排课 / `schedule`

- 所属实体：班次 → Slot[]
- 类型与单位：array；元素 {weekday, periods, rooms, weeks}
- 取值 / 示例：见下四行
- 来源：S2
- 质量与陷阱：**结构化的**（不是；分隔字符串）。实测：含排课 339 行、**空数组 2 行**（180089050200MB001Y-201 硕士学位英语（慕课学习）、180090125600PB001Y 工程伦理（玉泉路慕课））；每行条目数分布 0:2, 1:173, 2:127, 3:36, 4:3。

☒ 星期 / `schedule[].weekday`

- 所属实体：星期 (Weekday)
- 类型与单位：integer 1-7
- 取值 / 示例：1= 周一…7= 周日；分布 1:76, 2:92, 3:77, 4:75, 5:53, 6:88, 7:86
- 来源：S2
- 质量与陷阱：**1-based, 日 =7**（例：180080070100MX018Y 有 weekday 6 + weeks [[6,6]], 对应快照的“周六…第 6 周”）。映射到 一二三四五六日 时用 idx-1。

☒ 节次 / `schedule[].periods`

- 所属实体：节次区间 (PeriodRange)
- 类型与单位：array [first,last], 节 ∈ 1..13
- 取值 / 示例：[5,7]133、[10,12]108、[1,4]80、[3,4]50、[5,6]33
- 来源：S2
- 质量与陷阱：已是区间端点对，不是字符串；first==last 表示单节。

☒ 教室 / `schedule[].rooms`

- 所属实体：教室 (Room)
- 类型与单位：array of string
- 取值 / 示例：[" 教学楼阶一 3"]
- 来源：S2
- 质量与陷阱：数组（可多点）；与节次/周次**平行**：不含教室，域内无教室不变量。

☒ 周次 / `schedule[].weeks`

- 所属实体：周次区间集 (WeekRange set)
- 类型与单位：array of [start,end]
- 取值 / 示例：[[2,5],[7,17]]、单周 [[6,6]]
- 来源：S2
- 质量与陷阱：**区间集合，不是数字**；[6,6] 是退化为一点的集合。实测 min 周 =1, max 周 =22, 无 start>end。

主讲教师 / `teacher`

- 所属实体：班次
- 类型与单位：string
- 取值 / 示例：牛凌峰（空 69/341）
- 来源：S2
- 质量与陷阱：有缺失（慕课/讲座类）。

首席教授 / chief

- 所属实体：班次
- 类型与单位：string
- 取值 / 示例：示例常为空（空 202/341，占 59%）
- 来源：S2
- 质量与陷阱：缺失率极高，**不能当“该课没有负责人”的证据**，只能当“未登记”。

详情页 ID / course_id

- 所属实体：抓取句柄
- 类型与单位：string
- 取值 / 示例：315207
- 来源：S2
- 质量与陷阱：用于回抓 coursetime/{course_id}；非课程编码，不要与 code 混用。

开课校区 / campus

- 所属实体：校区 (Campus)
- 类型与单位：enum string
- 取值 / 示例：玉泉路 (341/341，空 0/341)
- 来源：S2
- 质量与陷阱：官方逐行真字段——修好 doc/data-authority.md 所说的老错觉（“公开库没有校区”）。但本文件已被 campus_code=20 过滤 → 也**无区分度**。

2c. 派生快照 course-library-snapshot.json (347 行, A2 冲突检查实际使用)

每行 = 一个班次；347 行 = 347 个不同 code（实测无重复）。**13 个字段实测 0 空值——“缺失”**在本文件里不表现为空串，而表现为哨兵值 (capacity="0")，或干脆没有该行。

课程编码 / code

- 所属实体：课程 + 班次
- 类型与单位：string
- 取值 / 示例：180080070100MX018Y；长度分布 18:256、20:7、21:70、22:1、23:13；第 18 位恒 Y
- 来源：派生快照（从教务系统爬取的课程数据快照；见 § 0.1）
- 质量与陷阱：班次后缀 -01/-W003/-304 等；去重须按前 18 位。

课程名称 / name

- 所属实体：课程

- 类型与单位: string
- 取值 / 示例: 实用最优化算法
- 来源: 派生快照
- 质量与陷阱: —

学分 / credit

- 所属实体: 学分
- 类型与单位: **number** (整型 291 / 浮点 56; 取值 {0.5,1,2,2.5,3,3.5})
- 取值 / 示例: 1、2.5
- 来源: 派生快照
- 质量与陷阱: 整型/浮点混用; 与 S2 字符串型不同。

课程属性 / attribute

- 所属实体: 课程属性
- 类型与单位: enum string
- 取值 / 示例: 学科核心课 77、公共必修课 76、专业课 74、专业核心课 53、公共选修课 50、研讨课 8、实验课 8、科学前沿讲座 1
- 来源: 派生快照
- 质量与陷阱: 与 S1 词表差在 实践课/科学前沿讲座 (§ 5.7)。

培养层次 / level

- 所属实体: 课程属性
- 类型与单位: enum string
- 取值 / 示例: 硕博通用课程 203、硕士课程 136、博士课程 8
- 来源: 派生快照
- 质量与陷阱: S1/S2-frozen 均无此字段; 只在这一源有。

所属学科 / subject

- 所属实体: 课程属性
- 类型与单位: string (77 个不同值)
- 取值 / 示例: 数学、语言学及应用语言学 30、工程管理 25
- 来源: 派生快照
- 质量与陷阱: S1/S2-frozen 均无; 粒度是” 一级学科/专业” 。

主讲教师 / teacher

- 所属实体: 班次
- 类型与单位: string
- 取值 / 示例: 牛凌峰 (空 0/347)
- 来源: 派生快照
- 质量与陷阱: 与 S2 不同: 这里 0 缺失 (S2 有 69 空)。

教室 / room

- 所属实体: 教室 (Room)

- 类型与单位: string (53 个不同值)
- 取值 / 示例: 教学楼阶一 3 (空 0/347)
- 来源: 派生快照
- 质量与陷阱: 单个字符串 (S2 是数组)。

星期节次 / time

- 所属实体: 星期节次 (DayPeriod)
- 类型与单位: string, ; 分隔段
- 取值 / 示例: 周三 (5-7); 周六 (5-7); 周一 (10-11); 周三 (10-11); 周六 (10-11)
- 来源: 派生快照
- 质量与陷阱: 与 **weeks 段数一一对应** (实测 0 处不对应)。段数分布: 1 段 177、2 段 128、3 段 40、4 段 2 (§ 5.2)。

开课周 / weeks

- 所属实体: 周次区间集
- 类型与单位: string, ; 分隔段; 段内 , 分隔区间
- 取值 / 示例: 第 2-5, 7-17 周; 第 6 周; 第 2-20 周; 第 2-5, 7-20 周; 第 6 周
- 来源: 派生快照
- 质量与陷阱: **区间集合, 不是数字** (第 6 周 = [(6, 6)])。段内区间数最多到 8; 实测无不可解析段。

容量 / capacity

- 所属实体: 容量
- 类型与单位: 非负整数字符串
- 取值 / 示例: 40、58、"0" 41 行 (空 0/347)
- 来源: 派生快照
- 质量与陷阱: 哨兵: "0" = 未设置 (§ 5.3); 真实 overcapacity (enrolled > cap > 0) 实测 0。

已选 / enrolled

- 所属实体: 已选
- 类型与单位: 非负整数字符串
- 取值 / 示例: 25; "0" 70 行 (空 0/347)
- 来源: 派生快照
- 质量与陷阱: enrolled=0 是真实“没人选”。

考试方式 / exam

- 所属实体: 考试方式
- 类型与单位: enum string
- 取值 / 示例: 闭卷笔试 110、课堂开卷 97、读书报告 55、大开卷 24、其它 (需说明) 51、文献综述 10 (空 0/347)
- 来源: 派生快照
- 质量与陷阱: 只此源有; 据 doc/data-authority.md 属“人工从 SEP 复制”的辅助字段, 不可当逐行官方口径。

3. S5 —培养方案（学分下限）

data/s5-programme/degree-requirements.json（校发培养字〔2025〕92号；085402 通信工程，2026 级）。

方案名 / programme

- 所属实体：培养方案
- 类型与单位：string
- 取值 / 示例：085402 通信工程（0854 电子信息专业学位，2026 级，集中教学在玉泉路 Y）
- 来源：S5
- 质量与陷阱：单一学生口径，不是全院规则。

总学分下限 / degree_track.master.total_min_credits

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：36
- 来源：S5
- 质量与陷阱：只给下限（I2 的 ceiling 半无来源）。

课程学习下限 / ...master.course_learning_min_credits

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：24
- 来源：S5
- 质量与陷阱：I2 主判据。

必修环节学分 / ...master.required_practicum_credits

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：12
- 来源：S5
- 质量与陷阱：非课程学分，不能与选课学分混算。

公共学位课下限 / ...master.public_degree_credits

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：7
- 来源：S5
- 质量与陷阱：与 categories[public_degree] 重复表达，改一处须同步。

专业学位课下限 / ...master.professional_degree_min_credits

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：12
- 来源：S5
- 质量与陷阱：同上 (= 核心课 4 + 专业课 4 …口径按类别表)。

转博（硕博）学分线 / `degree_track.combined_master_phd`

- 所属实体：培养方案
- 类型与单位：object
- 取值 / 示例：`course_learning_min_credits:32、public_degree_credits:11、professional_degree_min_credits:16 + note`
- 来源：S5
- 质量与陷阱：**陷阱**：转博**考核线**是 24（硕士课程学习），32 是转博后博士毕业前要求——note 明写，勿当考核线。

每学期学分下限 / `semester_min_credits.value`

- 所属实体：培养方案
- 类型与单位：integer 学分
- 取值 / 示例：`10;applies_to=" 秋季与春季每学期,不含人文讲座与科学前沿讲座" ;source=" 教务部选课说明"`
- 来源：S5
- 质量与陷阱：按“ 学期” 而不是“ 学年” ；`applies_to` 的排除项是实质约束。

类别 / `categories[]`

- 所属实体：培养方案
- 类型与单位：array of 7 objects
- 取值 / 示例：见 § 5.8
- 来源：S5
- 质量与陷阱：各类别 `kind` ∈ {degree, non_degree, non_course}；计分按类别独立核算（见 `known_modelling_gaps`）。

☒ 类别名 / `categories[].name + id`

- 所属实体：培养方案
- 类型与单位：enum
- 取值 / 示例：公共学位课/专业学位课 · 核心课/专业学位课 · 专业课/公共必修非学位课/公共选修课/专业选修课/必修环节
- 来源：S5
- 质量与陷阱：与数据源 `attribute` **不是**同一套词表——须经 `matches_attributes/matches_names` 映射。

☒ 下限 / `categories[].min_credits`

- 所属实体：培养方案
- 类型与单位：integer 学分

- 取值 / 示例: 7,4,4,1,2,2,12
- 来源: S5
- 质量与陷阱: 有的类别还有 min_courses (核心课 2、专业课 2)。

☒ 匹配规则 / matches_attributes matches_names also_allows_fixed_components

- 所属实体: 培养方案
- 类型与单位: string[] / object[]
- 取值 / 示例: 例: 核心课 matches_attributes:[" 学科核心课"," 专业核心课"]; fixed_components 用 name_contains 匹配课名
- 来源: S5
- 质量与陷阱: **按课名/属性软匹配**——课名改字即失配 (known_modelling_gaps[0] 记了 attribute 无法区分公共学位课与公共必修非学位课的缺口)。

☒ 固定组成 / fixed_components[].credits note

- 所属实体: 培养方案
- 类型与单位: 学分
- 取值 / 示例: 中特 2、自辩 1、学术规范 1 (通论 + 分论各 0.5)、英语 3; 实践 6/开题 2/中期 2/学术报告与社会实践 2
- 来源: S5
- 质量与陷阱: 含” 各 0.5 须合计 1” ” 免修免考记 EX 学分照得” 等**语义注记**, 不是可直接相加的行。

已知建模缺口 / known_modelling_gaps[]

- 所属实体: 培养方案
- 类型与单位: array (3 条)
- 取值 / 示例: 每条 {gap, resolution}
- 来源: S5
- 质量与陷阱: 是**诚实的未决项**, 不可当作已解决; 引用 S5 结论时须连同这 3 条一起引。

出处 / _source

- 所属实体: 溯源元数据
- 类型与单位: object
- 取值 / 示例: document/local_copy(~Desktop)/extracted:2026-09-27/extraction_method/note
- 来源: S5
- 质量与陷阱: note 明写” 未含院系加码” ——即**分母是校发口径**, 院系可更严。

4. 派生值对象与计算字段 (没有存储, 由上面字段推得)

周次区间 / WeekRange

- 所属实体: 值对象
- 类型与单位: (start,end) 整数周

- 定义 / 示例: (2,5); 第 6 周 $\rightarrow$ (6,6)
- 来源: 派生 (快照 / S2 的 weeks)
- 质量与陷阱: 区间半开/闭必须统一; [6,6] 不是“第 6 到第 6 周”以外的任何东西。

节次区间 / PeriodRange

- 所属实体: 值对象
- 类型与单位: (first,last), 节 $\in 1..13$
- 定义 / 示例: (5,7); 实测快照出现 18 种不同节次区间
- 来源: 派生
- 质量与陷阱: 节次上界因源而异 (S2 观测到 13)。

星期节次 / DayPeriod

- 所属实体: 值对象
- 类型与单位: (weekday, PeriodRange)
- 定义 / 示例: (3, (5,7)) = 周三第 5-7 节
- 来源: 派生
- 质量与陷阱: 快照用字符串、S2 用 weekday+periods——同一概念两种编码, join 要归一化。

时段 / Slot

- 所属实体: 值对象
- 类型与单位: DayPeriod + WeekRange set
- 定义 / 示例: 切分单位, 冲突规则比较的对象
- 来源: 派生
- 质量与陷阱: 一个班次可有多个 Slot (快照里靠; 分段表达)。

冲突 / Conflict

- 所属实体: 值对象 (派生布尔)
- 类型与单位: boolean
- 定义 / 示例: weekday 相同 $\wedge$ 节次相交 $\wedge$ 周次相交
- 来源: 派生
- 质量与陷阱: **从不存储**; 只比“星期 + 节次”会误报 (半学期同格), 不看周次会漏报 (domain-glossary §3、invariants I1)。

满 / SectionFull

- 所属实体: 班次状态
- 类型与单位: boolean
- 定义 / 示例: enrolled = capacity (且 capacity 已知)
- 来源: 派生
- 质量与陷阱: capacity 未知 (空/0//) 时该状态**不可判定**, 不是“未滿”。

超容量 / SectionOverCapacity

- 所属实体: 班次状态
- 类型与单位: boolean

- 定义 / 示例: `enrolled > capacity > 0`
- 来源: 派生
- 质量与陷阱: 实测这份数据里 0 例 (§ 5.3); `capacity=0`// ‘ 的行不算, 须单独报告。

5. 实测统计 (本会话所有关键数字与出典)

方法: python3 内联脚本, `json.load` 后按行统计; 无临时文件。下面每行「出典」给出可复算的 python 表达式 (变量名: `s1=load("…sections-snapshot.json")["courses"]`、`s2=load("…/89576-campus20.json")["rows"]`、`snap=load("…course-library-snapshot.json")["courses"]`, `base=lambda c:c[:18]`)。

5.1 ① 条数与去重后课程编码数

S1

- 记录数: 732
- 去重 code 数 (原样): 414
- 去重后基码数 (前 18 位): **413**
- 出典: `len(s1), len({c['code']}), len({base(c['code'])})`

S2 (89576 原始)

- 记录数: 341
- 去重 code 数 (原样): 341
- 去重后基码数 (前 18 位): **270 / 282**
- 出典: `len(s2)`, 两种去后缀口径见下

派生快照

- 记录数: 347
- 去重 code 数 (原样): 347
- 去重后基码数 (前 18 位): **277**
- 出典: `len(snap)`, 同上
- S1 编码长度分布 {18:731, 22:1} (唯一 22 位 = 180089050200MB001Y-201)。
- S2 编码长度分布 {18:249, 20:7, 21:70, 22:2, 23:13}; 快照 {18:256, 20:7, 21:70, 22:1, 23:13}。
- **第 18 位字符**: 三源实测 100% 为 Y (S1 732/732、S2 341/341、快照 347/347) → `campus_rule` 成立。

5.2 ② 时间字段分段结构

指标	值	出典
快照 time 空 / weeks 空	0 / 0 (347)	<code>sum(1 for c in snap if not c['time'].strip())</code>
time 段数分布 (切)	{1:177, 2:128, 3:40, 4:2}	<code>Counter(len(c['time'].split(';')))</code>
weeks 段数分布	{1:177, 2:128, 3:40, 4:2}	<code>Counter(len(c['weeks'].split(';')))</code>
(与 time 完全相同)		
time/weeks 段数不一致的行	0	逐行比较两式
不可解析的周次段	0	正则 <code>^ 第 [\d,\-]+ 周 \$</code>
weeks 每段内区间个数分布	{1:324, 2:211, 3:13, 4:10, 5:1, 7:1, 8:1}	<code>Counter(len(seg.split(',')))</code>
星期 token 计数	一 79 二 96 三 81 四 75 五 52 六 90 日 88	正则 <code>^ 周 ([一二三四五六日天])</code>
节次区间 (快照) 不同取值数	18; top (5,7)139、(10,12)109、(1,4)80、(3,4)50	<code>Counter(...)</code>
S2 结构化: 含排课行 / 空排课行	339 / 2	<code>sum(bool(r['schedule']))</code>
S2 每行排课条目数分布	{0:2, 1:173, 2:127, 3:36, 4:3}	<code>Counter(len(r['schedule']))</code>
S2 weekday 分布 (1-7)	1:76 2:92 3:77 4:75 5:53 6:88 7:86	<code>Counter(e['weekday'])</code>
S2 weeks 全局 min/max	1 / 22; start>end 0 例	<code>min/max(...)</code>
S2 每段周次区间数分布	{1:314, 2:205, 3:15, 4:10, 5:1, 7:1, 8:1}	<code>Counter(len(e['weeks']))</code>

5.3 ③ capacity / enrolled 缺失值与 0 值

快照 347

- capacity 空: 0
- capacity = 0 (快照) / = "/" (S2): 41 (字面 "0")
- capacity > 0: 306
- enrolled 空: 0
- enrolled = 0: 70
- enrolled > 0: 277

S1 732

- capacity 空: 732 (全空)
- capacity = 0 (快照) / = "/" (S2): 0
- capacity > 0: 0

- enrolled 空: **732** (全空)
- enrolled = 0: 0
- enrolled > 0: 0

S2 341

- capacity 空: 0
- capacity = 0 (快照) / = "/" (S2): **40** (字面 "/", 另 0 值 0 条)
- capacity > 0: 301
- enrolled 空: 0
- enrolled = 0: 75
- enrolled > 0: 266
- 快照 capacity=="0" 且 enrolled>0 的行 = **40** (与 invariant-checks.md §3 一致, 本次复核相同); 剩 1 行 capacity=="0" 且 enrolled==0 (180101120500PB003Y-03 学术道德与学术写作规范)。
- 快照里 enrolled > capacity > 0 的**真实超容量** = 0。
- 出典: `sum(1 for c in snap if c['capacity']=='0')` 等; `sum(1 for r in s2 if r['capacity']=='/')=40`。

5.4 ④ 周次超出学期长度

- 参照: 秋季学期 **20 教学周**(doc/prototype-issues.md P15、scripts/build_web_prototype.py 明写“ 秋季学期共 20 教学周”)。
- 快照: **周次上界 > 20 周的行 = 16** (全部恰好到第 **22** 周, 全部是 初级汉语 1/初级汉语 2, 编码 180101050102PB001Y-*(5 条) 与 180101050102PB002Y-*(11 条), 属性均为 公共必修课)。
- 快照每门最大周次直方图: {7:2, 8:3, 9:12, 10:13, 11:6, 12:16, 13:31, 14:17, 15:34, 16:45, 17:25, 18:32, 19:29, 20:66, 22:16} (无 21 周)。
- 阈值敏感性: >18 周 **111** 行、>19 周 **82** 行、>20 周 **16** 行、>22 周 **0** 行。
- S2 原始同样 **16** 行 (同一批汉语课)。
- 出典: `sum(1 for c in snap if max(int(x) for x in re.findall(r'\d+', c['weeks']))>20)`。
- **【注意】诚实边界**: 三个冻结源里**没有任何字段声明学期长度**; “20 周” 是 prototype-issues.md 引用的教务口径, 不是数据。这个数字随学期口径变化, 引用时必须带口径。

5.5 ⑤ 同一课程同一时段出现在多个周次块

- 快照中“ 某课程的同名 (星期, 节次) 段在 ; 分段中出现 ≥ 2 次” 的行 = **4** (与 invariant-checks.md §0 一致):

课程编码	课程	time	weeks
180086081201P3003Y	超大规模集成电路基础	周三 (5-7); 周三 (5-7); 周六 (5-7)	第 3-5,7-15 周; 第 16 周; 第 6 周
180096120400MX020Y	科技创新政策分析	周一 (10-12); 周一 (10-12)	第 2-3,5-12 周; 第 13 周
180202085403P2004Y	集成电路制造技术	周一 (10-12); 周一 (10-12); 周六 (1-3)	第 3-13 周; 第 14 周; 第 2-5,7-13 周
180202085403PB001Y	学术道德与学术写作规范-分论	周三 (10-11); 周三 (10-11)	第 3-5,7 周; 第 8 周

- 同一课程” 整段 (time 且 weeks) 完全重复” 的行 = 0 → 这 4 行是**半学期拆分** (同一格、不同周次), 不是数据重复。正是 I1 要保护的场景。
- S2 原始: 同一行内 (weekday, periods) 出现在 >1 个 schedule 条目 = 0 (它那 4 行在原始里是**不同周次**的条目, 被快照压平成重复段)。
- 出典: Counter(re.sub(r'\s', '', s) for s in c['time'].split('; '))。

5.6 S1 time/weeks 为空 (A1 复核)

- S1 顶层 time 非空 = 0/732; weeks 非空 = 0/732。
- sessions[] 的 time/weeks/room 非空 = 0; sessions 长度 = 732/732 恒为 1。
- 结论: A1 ” 0/732” **本次复核成立**, 且连退路 (sessions) 也是空的。

5.7 词表与跨源一致性 (; 段数之外的对账)

- 属性词表: S1 有 实践课 (1) 无 科学前沿讲座; 快照/S2 有 科学前沿讲座 (1) 无 实践课。
- 快照 vs S2 原始, 在 **251 个可 1:1 对应的共享基础**上:
 - 星期节次集合不一致 **8 例** (占 3.2%); 例如 1801010705I0P1003Y 快照 周一 (9-11) / S2 周一 (10-12)。
 - 周次整数集合不一致 **9 例** (占 3.6%); 例如 180086081202PX005Y 快照含第 5 周、S2 不含。
- **计数口径 (必须写在数字旁边)**: S2 的 341 行按「去掉数字班次后缀 -NN」去重 = **282** (A1 简报 E3 用的口径); 按「去掉第一个 - 之后的全部后缀」/ 取前 18 位 = **270**。两者差 12 个字母后缀 (-S503、-W501、-201、-304 等)。下表统一用 **270 (更激进)**; 引用 282 时说明是「仅去数字后缀」口径。
- 基码重叠: S1 413、S2 270 (282 口径下为 282)、快照 277; S1 ∩ S2raw **224**、S1 ∩ 快照 **230**、快照 ∩ S1 **47**、S2raw ∩ 快照 **269**。
- 与 doc/data-authority.md ” [表情] 与 [表情] 对同一门课的星期 + 节次 + 周次**完全一致**” 相比: 在被引用的那个样例上一致, 在全集上约 **96-97% 一致**——已不是” 完全一致” (§6 陷阱 T8)。

5.8 分类分布 (供字典取值列复核)

字段	分布	出典
S1 semester	秋季 400 / 春季 332	Counter
S1 department	国际学院 160、计算机科学与技术学院 87、电气与通信工程学院 87、外语系 73、网络空间安全学院 49、集成电路学院 41	Counter().most_common
S1 source	开课计划表 0828.xlsx 441 / 核心课和专业课列表 0828.xlsx 291	Counter
快照 level	硕博通用课程 203 / 硕士课程 136 / 博士课程 8	Counter
快照 exam	闭卷笔试 110 / 课堂开卷 97 / 读书报告 55 / 大开卷 24 / 其它（需说明）51 / 文献综述 10	Counter
快照 subject	77 个不同值; top 语言学及应用语言学 30、工程管理 25、外国语言文学 19	Counter
快照 room	53 个不同值	set
S2 hours	4099、6057、3028、3627、5024、12822、3221、2017	Counter
S5 categories	7 类（含 min_credits 7/4/4/1/2/2/12; 核心课与专业课另有 min_courses=2）	读文件

6. 陷阱清单——逐条与” 示例陷阱” 对账（含不符发现）

#	示例/文档中的说法	实测结论	判定
T1	capacity=0 是” 未设置” 哨兵而非真实容量	快照 41 行 capacity=="0", 其中 40 行 enrolled>0; 真实 enrolled>capacity>0 = 0	✓ 在 347 快照上成立
T2	同上（陷阱的措辞）	** 官方 S2 原始里” 未设置” 哨兵是 "/" (40 行), 不是 0; 0 值 0 条。** 快照把 "/" 渲染成 0	【注意】〔表情〕部分不符（口径差异，不是矛盾）：官方原文用 "/"；快照把 "/" 渲染成 0。判据不变——两者都是「未设置」哨兵
T3	time/weeks 用；分段且两字段段数对应	段数分布完全相同，不一致 0 行	✓ 成立（仅对 347 快照；S1 两列全空，S2 是结构化数组，不存在” 段数” 概念）
T4	周次是区间集合不是单个数字	第 2-5, 7-17 周 可展开为整数集合；第 6 周 = [(6,6)]; start>end 0 例	✓ 成立

#	示例/文档中的说法	实测结论	判定
T5	同一课程编码可有多班次后缀	S1 732→414→ 413 ; S2 341→ 282 (只去数字后缀) / 270 (去全部后缀); 快照 347→ 277	✓ 成立 (口径见 §5.7)
T6	S1 里 time/weeks 为空的行数 (A1 报 0/732)	顶层 0/732 ; sessions[] 也 0; sessions 恒 1 条且空	✓ 成立 (A1 复核通过)
T7	学期字段不在开课计划表内	不符 : committed 的 S1 快照 每行都有 semester (秋季 400 / 春季 332), 另有 campus(玉泉路)、department、source。doc/data-authority.md 更把〔表情〕工作簿列为“开课学期”的 权威源 (★ 只有它有)	× 与数据不符 : S1 有学期; 真正 没有 逐行学期的是 S2 与快照 (S2 只能按 term_id/文件推)
T8	doc/ data-authority.md: “〔表情〕(官方) 与 〔表情〕(派生快照) 对同一门课的星期 + 节次 + 周次 完全一致 ”	在 251 个共享 1:1 基础上: 星期节次不一致 8 例 (3.2%)、周次集合不一致 9 例 (3.6%)	【注意】〔表情〕 不符 (规模上): 样例一致, 全集约 96-97%, 非” 完全一致”
T9	S2 = 19 字段原始抓取	committed 89576-campus20.json 每行 15 个 key ; 19 字段是上游未入库原始下载的表头 (live-listing-provenance.md)。committed 文件不含 level/exam/subject/授课方式/是否远程教学/助教	【注意】〔表情〕 措辞须区分 : 19 字段 ≠ 本文件
T10	两个”冻结数据源” = S1 + S2	另有一份 347 快照是 派生数据 (从教务系统爬取, 归档渠道见 live-listing-provenance.md), 与 S2 不是同一次抓取, 二者在 251 个共同基础上仍有 3% 级分歧 (T8)	【注意】〔表情〕 命名分歧 : A2 冲突检查实际跑在这份派生快照上, 引用其数字时须标清上游
T11	(领域) 教室/考试方式	快照 room 0 空、exam 6 值; S2 rooms 是数组、 无 exam ; S1 exam 全空	提示: 三源字段覆盖不同, 别跨源假定字段都在

#	示例/文档中的说法	实测结论	判定
T12	(领域) ” 课程满” 是班次状态	快照真实 enrolled>capacity>0 = 0 例; capacity 未知 (空/0//) 时” 满/未满” 不可判定	提示: capacity 缺失率高时不可下” 无超容量” 结论
T13	缺失 ≠ 空	快照 13 字段 0 空值 ; ” 缺” 表现为哨兵 0// 或整行缺席。S2 chief 空 202/341(59%)、teacher 空 69/341	提示: 别用” 空串” 作唯一缺失判据
T14	编码” 不是不透明的”	前 18 位为基码, 第 18 位恒 Y (三源 100%); 长度 18/20/21/22/23	成立; 但 只能 推校区, 不能推学期/容量

7. 未证实 / 需外部依据 (不编造)

- **学期长度**: 三个冻结源无此字段; ” 秋季 20 周” 来自 doc/prototype-issues.md 引用, **非数据** (§ 5.4)。
- **S2 原始文件的确切逐字段上游映射**: live-listing-provenance.md 给出 19 个 SEP 表头; committed 15 key 与 它们的**逐列一一对应**我只核到 13 个语义等价列 (另 2 个 course_id/schedule_pages_sha256/campus 为抓取侧新增), 其余为**结构推断**, 文中已标注, 未逐条证实。
- **department 是否等同工作簿” 课程所属学科”**: sections-snapshot.json 的取值 (如” 国际学院” ” 计算机科学与技术学院”) 读起来是**开课院系**而非学科; 与 data-authority.md 所述工作簿列” 课程所属学科” 是否同一列, **未证实**。
- **S1 的 semester 是工作簿列还是快照生成时按工作表名回填**: snapshot-provenance.md 的导入提示只列了” 缺少 培养层次/开课周/星期节次”, 未提学期; 因此两读法都可能, **未证实** (但 committed 文件里字段确实存在)。
- **S2 capacity 数值的官方口径** (限选是否为” 计划容量” 或” 实际可容”): 数据无说明, **未证实**。

实测于本会话 (2026-10-04), 全部为 inline *python3* 对 committed 文件运行; 哈希见 § 0.2。未新增任何脚本文件。

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A2 —Invariants and their automated checks

Three invariants, each stated in the domain's vocabulary (see domain-glossary.md), each backed by a check that runs on the committed data. Raw output: data/derived/a2/invariant-checks.md. Re-run:

```
python3 scripts/a2_check_invariants.py           # writes the raw output, exits 0 whether
```

A violation found by a check is a **finding about the data**, not a script error. Two of the three invariants below are violated by the real data, and those violations are the interesting part.

I1 —No timetable may contain two sections whose slots overlap

Statement. For any two sections a, b in the same plan:

```
conflict(a, b) ⇔ weekday(a) = weekday(b) ∧ periods(a) ∩ periods(b) ≠ ∅ ∧ weeks(a) ∩ weeks(b) ≠ ∅
```

All three conditions at once. Any two of them alone do not make a clash (the full truth table is in conflict-decision-table.md).

Why it is worded this way. The naive screen —same weekday and overlapping periods —is wrong in both directions: it **over-reports** (two courses in the same slot at disjoint week ranges are perfectly compatible, which is the normal half-semester pattern in this data) and it **under-reports** nothing, because it is a superset condition. A second naive screen —comparing whole-semester slots —**under-reports** the case of two courses that only share part of a slot's week range.

Check. scripts/a2_check_invariants.py section 1-2:

reading	result
decision-table cases (6 canonical cases incl. the week-interval trap)	all behave as specified
catalogue as a whole (347 scheduled courses, 146,427 segment comparisons)	4,757 conflicting course pairs = 7.9 % of course pairs = the conflict density of the offer set
an illustrative plan of 6 courses (data/derived/a2/sample-plan.json)	1 internal conflicting pair
the three courses named in A1 (计算机代数 / 数字图像处理与分析 / 深度学习)	all three at 周四 第 10-12 节 → 3 conflicting pairs

The A1 case, reproduced. A1 asserted that three courses in the author's plan all sit in one slot. That claim is re-derivable from committed data alone:

course	code	slot	weeks
计算机代数在科学与工程中的应用	180081070100PX001Y	周四 第 10-12 节	2-18
数字图像处理与分析	180093081002P3008Y	周四 第 10-12 节	2-19
深度学习方法与应用	180206085406P3002Y	周四 第 10-12 节	2-4, 6-15

All three pairs conflict. The clash is a property of the **offer set**, which is why the planner needs the rule rather than a printed plan.

Limit. The catalogue-level figure (4,757) is not a violation of I1: a timetable is a chosen subset, and a student cannot take every course. Only the plan-level reading is an invariant check proper.

I2 —Planned credits must satisfy the programme's floors

Statement. For the planned course set P and the programme's floors f : $\sum \text{credits}(P) \geq f$ for each floor that the programme states.

Check. `scripts/a2_check_invariants.py` section 4: reads `data/s5-programme/degree-requirements.json` (master track: total 36, course learning **24**, public degree 7, professional degree 12, practicum 12) and the plan under test.

reading	result
illustrative plan (6 courses)	13.5 credits → floor not satisfied (the check works; the plan is deliberately a sample)
the author's real plan (private; only its aggregate is committed)	course-learning credits reached 26 (floor 24) —see <code>data/derived/credit-gap-summary.md</code>

Honest limits. (a) The programme document states **floors only**, so the ceiling half of the invariant has no source and is left unimplemented rather than invented. (b) The per-course plan is personal data and is not committed, so the check ships with a committed illustrative plan so that it is runnable from the repository alone; the real plan's aggregate is quoted from the committed summary. (c) The "planned course missing from the catalogue" branch is a real case in 春季 and is not exercised by the illustrative plan.

I3 —An enrolled count must be interpretable against a known capacity

Statement (as first drafted). $\text{enrolled} \leq \text{capacity}$ for every section.

Statement (as the data forced it to be). enrolled must be interpretable against a **known** capacity; capacity ∈ {empty, "0"} means **unknown**, and unknown capacity must be surfaced to the planner —never silently read as " unlimited" and never silently read as " full" .

Where the sentinel comes from. In the official listing the unset value is written "/" (40 rows); the committed derived snapshot renders that as 0. Either way it means not set —but it is worth knowing which layer produced the nicer-looking number (see data-dictionary.md § 6 T2).

Why the statement changed. The first run reported 40 violations. Every single one of them had capacity = 0 with a non-zero enrolment —a row with 86 students against capacity 0 is not a section that admits nobody: it is a row where the capacity field was never filled in. Reading it literally would have produced a headline finding (" the school oversells 40 sections") that is simply wrong. After separating the sentinel:

reading	result
sections with both figures present	347
capacity missing/empty	0
rows with capacity 0 and enrolled > 0 (sentinel = not set)	40
violations with a real capacity (enrolled > capacity > 0)	0

Check. scripts/a2_check_invariants.py section 3 —it reports the two groups separately, so a sentinel row can never be laundered into a violation count.

Limit. With capacity 0 meaning " not set" , the honest conclusion about over-capacity in this snapshot is " cannot be determined from this field" , not " none exists" . A1 observed over-capacity from an earlier crawl (422/420 for 自然辩证法 sections); the two observations differ because they were taken at different times, which is exactly why every frozen input in this repository carries a capture time and a hash.

What the three invariants do not cover

- room clashes (same room, overlapping slots) —the data supports it, no invariant claimed;
- teaching loads, exam timetables, and travel time between campuses;
- fairness of the grab order (first-come-first-served is taken as given, not modelled);
- anything about the registration transaction itself, which is deliberately out of scope.

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A2 —Material rule as a decision table, contract and temporal property

A2 asks for one material rule expressed as a decision table, a contract, an assertion, or a temporal property. This repository does it three ways for the same rule, because the three forms disagree in the way they fail.

The rule: **I1** —**two sections may not overlap in a timetable.**

1. Decision table (the primary form)

Inputs (all three are conditions on a pair of sections): same weekday?, period ranges overlap?, week ranges overlap?

1

- same weekday: N
- periods overlap: —
- weeks overlap: —
- Verdict: **no clash**
- Why: different days cannot clash

2

- same weekday: Y
- periods overlap: N
- weeks overlap: —
- Verdict: **no clash**
- Why: same day, different periods

3

- same weekday: Y
- periods overlap: Y
- weeks overlap: **N**
- Verdict: **no clash**
- Why: same slot in disjoint week blocks —the half-semester pattern

4

- same weekday: Y
- periods overlap: Y
- weeks overlap: **Y**
- Verdict: **CLASH**
- Why: the only clashing row

The two conditions marked — are don't care: their value cannot change the verdict. That is the whole point of the table —a naive screen treats " same weekday $\wedge$ periods overlap" as sufficient, which is row 3 stated wrongly as a clash, and a whole-semester screen hides row 4 when the overlap is only partial.

Executable form. `scripts/a2_check_invariants.py` implements exactly this table:

```
def conflict(a, b):
    return (a.weekday == b.weekday
            and overlap_periods(a, b)
            and overlap_weeks(a.weeks, b.weeks))
```

Its canonical cases (printed into `data/derived/a2/invariant-checks.md` §1, all four rows plus two registry cases) —all six behave as the table says:

case	verdict
same weekday + overlapping periods + overlapping weeks	clash
weeks disjoint [3,5] vs [6,8]	no clash (row 3)
weeks interleaving [3,5] vs [4,8]	clash (row 4) —the trap
same weeks, periods apart	no clash (row 2)
same weeks and periods, different weekday	no clash (row 1)
identical section (the same section compared with itself)	clash —matters because the calling code must never
compare a section with itself	

2. Contract form (design by contract)

operation `detectClash(a: Section, b: Section) -> Conflict | NoConflict`

```
pre:      a ≠ b                                # a section never clashes with itself
          a.weekday, a.periods, a.weeks are parsed # an unparsable field is an error, not "
post:      result = Clash ⇐ a.weekday = b.weekday
          ∧ a.periods n b.periods ≠ ∅
          ∧ a.weeks n b.weeks ≠ ∅
```

invariant (of the caller): collected plan $P \sqsubseteq \forall a, b \in P, a \neq b : \text{detectClash}(a, b) = \text{NoConflict}$

The two pre-conditions are the ones that actually bite:

- **$a \neq b$** —a section compared with itself always "clashes", so a caller that forgets the guard reports the whole catalogue as broken. (The check counts self-clashes separately; the snapshot has 0 because the data contains no exact duplicate segments —but the guard is what makes that a finding rather than luck.)

- **fields must be parsed, not defaulted** —an empty or unparseable time field must raise, never be silently read as " no clash" . A default of " no clash" turns missing data into a false all-clear, which is the single most dangerous failure mode for this whole project.
-

3. Temporal-property form

For the registration window, where the plan changes over time:

$G(\text{selected}(a) \wedge \text{selected}(b) \wedge a \neq b \rightarrow \neg \text{overlap}(a, b))$

" At every moment, if two distinct sections are selected, they do not overlap." Two refinements that the model makes explicit (see `domain-glossary.md` for the events):

- $F(\text{SectionFull}(s) \wedge \text{selected}(s))$ —a section that was selected may become full later: the property holds on selection state, not on capacity state, so a delayed clash is possible in reality and is surfaced as a `SectionFull` event for the plan, not hidden;
 - $F(\text{PlanRevised}) \rightarrow G(\neg \text{overlap})$ —after every plan revision the property must be re-established; that is why the product re-checks the whole plan after an edit instead of checking the edited pair only.
-

4. What this rule does not decide

- two sections in the same slot with **non-overlapping** week ranges are allowed, by design (row 3) —this is therefore not a " slot is taken" check but a " time is taken" check;
- a clash is derived, never stored: there is no `conflicts_with` field anywhere in the data, so a stale stored answer is impossible by construction;
- room, campus-travel and exam clashes are out of scope (see `invariants.md`, final section).

\newpage

A2 —Data governance: source, authority, quality, retention, deletion, privacy

This document answers one question at field granularity, not at " dataset" granularity: **for each field of each source, where does the value come from, who is allowed to believe it, how long it is kept, how one person's copy of it is deleted, and what it costs if the field is wrong.**

Written 2026-10-05. Companions: `domain-glossary.md` (entities and value objects), `invariants.md` (the three rules and their limits), `conflict-decision-table.md` (the 8-row

rule), data-dictionary.md (field types), model-walkthrough.md (which cites this file's § 5 as the executable deletion procedure).

Source numbering is fixed and must not be renumbered. S1 = the official course-planning workbook (2026-08-28); S2 = the university's public course listing, jwbaucas.ac.cn (crawled 2026-09-27); S3 = the author's own two saved plan versions; S4 = the AI-generated plan, which is **derived** from S1 and therefore is not an independent source; S5 = the programme requirements (校发培养字〔2025〕92号); S6 = the school's autumn course-selection notice. Two further inputs exist that are **outside the S1-S6 numbering** and are labelled X1 / X2 below so that they can never be mistaken for an official source.

1. Source inventory

Acquisition time, method, hash, and authority grade, per source. Authority grade is one of three words and is read per-field in § 2:

- **authoritative (权威)** —this field is taken from here;
- **reference (参考)** —read for context, or used when the authoritative source lacks the field, and labelled as such in the output;
- **cross-check only (仅交叉验证)** —never a value's origin; used only to confirm or to distrust an authoritative value.

S1

- Source (form): Official course-planning workbook —2026–2027 学年秋季和春季开课计划表 0828.xlsx + …研究生核心课和专业课列表…0828.xlsx (xlsx, 2 workbooks)
- State: planned
- Acquired: issued 2026-08-28; imported & frozen **2026-09-27**
- Method: python3 scripts/freeze_snapshot.py (structured import via the planner backup)
- Artefact + sha256: data/s1-plan-workbook/sections-snapshot.json ffcf1ae023e59fe8a09d8 (+ .sha256); provenance snapshot-provenance.md
- Authority grade: **authoritative for 开课学期**; reference for 课程属性/学分/学时/开课校区

S2

- Source (form): Public course listing, https://jwbaucas.ac.cn/sc/public/coursePublic (list) + /sc/course/coursetime/{cid} (schedule) —server-rendered HTML, no login, no captcha
- State: current at capture
- Acquired: 2026-09-27 (2026-27 terms captured **16:31:49** Asia/Shanghai)
- Method: python3 scripts/fetch_official_db.py --terms … --campus 20; --sleep 0.25, 3 retries; cached
- Artefact + sha256: data/s2-public-listing/89576-campus20.json (341 rows) 6510aa2e4127a096208de7d02916c6bc5c8643f66f050d68470cff2d83362db5 (+

.sha256); 6 past terms under history/ with per-term list-page and output hashes (e.g. 74468 → a90296c9..., 84068 → 6e1f4505...); provenance official-db-provenance.md

- Authority grade: **authoritative for** 开课校区, 课程属性, 学分, 课时, 开课周, 星期节次, 上课地点, 限选, 已选, 教师; also the sole source of 学期 boundaries only as " which term it was crawled under"

S3

- Source (form): The author's two saved plan versions —选课规划-已选课程-2026-09-05.xlsx, ...-2026-09-15.xlsx (xlsx, **not committed**)
- State: own working record
- Acquired: exported 2026-09-05 11:33 / 2026-09-15 21:41; inspected **2026-09-27 15:37**
- Method: python3 scripts/inspect_plan_versions.py; hash + cross-version cell diff
- Artefact + sha256: only the report is committed: data/s3-my-plan-versions/plan-versions.md; the two files' hashes are recorded there (8d6f9e65..., dcc6cae7...)
- Authority grade: **authoritative for " what I actually chose / what changed"** ; nothing else

S4

- Source (form): AI-generated plan (2026-08-28)
- State: derived from S1
- Acquired: 2026-08-28
- Method: —
- Artefact + sha256: ai/ai-use-log.md, doc/prototype-issues.md
- Authority grade: **not a source** —kept only as a failure case; excluded from the independence count

S5

- Source (form): Programme requirements —中国科学院大学电子信息专业学位研究生培养方案（校发培养字（2025）92号）(PDF, not committed)
- State: official, stable
- Acquired: extracted **2026-09-27**
- Method: verbatim text extraction, § 第二部分 + 六、必修环节及学分要求
- Artefact + sha256: data/s5-programme/degree-requirements.json + .sha256 sibling (added 2026-10-04); provenance data/s5-programme/provenance.md —but see § 3.5, the transcription itself is still manual
- Authority grade: **authoritative and sole source for the credit floors and degree-course rules**

S6

- Source (form): School notice on autumn course selection
- State: official, event-scoped

- Acquired: read 2026-09
- Method: quoted text
- Artefact + sha256: cited in the A1 brief and doc/evidence-`ledger.md`; no archived copy
- Authority grade: **authoritative for the registration window and the conduct rules** (no cross-campus for centralised-teaching students; no disrupting/hoarding registrations)

X1

- Source (form): Third-party course library, `courseplanner.cysdy.cn/default-courses.json` (JSON)
- State: snapshot, maintainer-updated 2026-09-14 13:00
- Acquired: raw download 8,103,455 bytes; **not committed**; hash re-verified 2026-09-27
- Method: `curl + python3 scripts/fetch_public_listing.py`
- Artefact + sha256: raw `7c6090e40801d1ba4ebf3665c54eafaf4ec6e5e3565d996df74e63fac53e84` (kept, not committed); frozen subset `data/derived/course-library-snapshot.json` (347 Yuquanlu codes) `3724ade7ff05dcd3465c017f965feffe0eaa26cfbcaafebf976d268003122a` provenance `data/derived/live-listing-provenance.md`
- Authority grade: **reference and cross-check only** —never authoritative for any field

X2

- Source (form): SEP auxiliary fields, frozen from X1's raw download
- State: derived from X1
- Acquired: frozen 2026-09-27
- Method: `python3 scripts/freeze_aux_fields.py` (refuses to run unless the raw download's sha256 prefix matches `7c6090e4...`)
- Artefact + sha256: `data/derived/sep-aux-fields.json` (2,951 course codes) `7d48eb75e1af1477d2475d710449f2065307d36eb0658a615cea48fb906061cc`
- Authority grade: **reference only** (授课方式/考试方式/是否远程教学/助教/培养层次)

Independence, stated honestly (unchanged from the ledger). S1, S2, S5, S6 are official publications —one type. S3 is the author's own record —the other type, and the only record of what actually changed between two planning attempts. S4 is derived from S1 and is not counted. X1 is a third-party re-publication of SEP data: it is a **different pipeline over the same underlying facts**, which is why it is usable as a cross-check (S2 and X1 agree exactly on 星期节次 + 开课周 for sampled courses, e.g. 实用最优化算法 = 周三第 5-7 节, weeks 2-5/7-17) and not usable as an authority.

1.1 What is deliberately not in the repository

Repeated here because retention (§ 4) and deletion (§ 5) depend on it: student ID, name, programme code and the personal plan spreadsheets; any other student's selection record (never read); per-course personal priority detail and the personal credit-gap file (`data/private/`); the prototype HTML (it embeds the personal plan); the raw HTML crawl cache

(data/s2-public-listing/cache/); the 8.1 MB X1 raw download; the 4,877-record planner backup; and credentials/tokens/cookies, of which there are none anywhere in history.

2. Field-level authority

One row per field. " Authoritative source" is where the value is taken from; " fallback" is what is allowed when the authoritative source has no value; " conflict rule" is what happens when two sources disagree.

课程编码 code

- Type: 18-char string
- Authoritative: S2 (live)
- Fallback (labelled in output): S1 (planned)
- Conflict rule: Treat as identity: if the codes differ the rows are **different courses**, never merged. If S1 and S2 carry the same code with different names, S2 wins for the name and the difference is logged.

课程名称 name

- Type: string
- Authoritative: S2
- Fallback (labelled in output): S1
- Conflict rule: S2 wins (S2 is the current system); a mismatch is logged as a naming drift, not silently overwritten.

开课校区 campus

- Type: enum (Y/H/Z by code position 18; S2 carries the word)
- Authoritative: **S2** (a real column)
- Fallback (labelled in output): S1 (real column)
- Conflict rule: S2 wins. For **X1 only** —which has **no campus field at all** —campus is inferred from code position 18 and must stay labelled inferred; it is never promoted to a fact.

开课学期 semester

- Type: enum (秋/春/both)
- Authoritative: **S1** —the only source that has it (208 autumn-only, 179 spring-only, 27 both in the kept subset)
- Fallback (labelled in output): none
- Conflict rule: S2 has no per-course term and X1 has none; the only S2-derived term statement is " this row belongs to the term it was crawled under" (2026-27 秋 = termId 89576). If S1 says 春季 and S2 shows it in the autumn crawl, that is a **coverage finding**, reported, not resolved.

课程属性 **attribute**

- Type: enum
- Authoritative: **S2**
- Fallback (labelled in output): S1
- Conflict rule: S2 wins; S1 is the corroboration (data-authority § 二 reached the same conclusion: 〔表情〕 may mutually confirm, 〔表情〕 is authoritative).

学分 **credit**

- Type: number
- Authoritative: **S2**
- Fallback (labelled in output): S1
- Conflict rule: S2 wins. Type caveat: S2 stores "1.00" (string), S1 stores a number — coerce at the boundary, keep the source string.

学时 **hours**

- Type: number
- Authoritative: **S2**
- Fallback (labelled in output): S1
- Conflict rule: S2 wins.

开课周 **weeks**

- Type: **interval set**, e.g. [(2,5),(7,17)]
- Authoritative: **S2** (detail page) —347/347 rows in the frozen subset
- Fallback (labelled in output): none
- Conflict rule: S1 has this field in **0 of 732** rows; X2/X1 are cross-check only. Never a single number (see glossary: 第 6 周 = [(6,6)]).

星期节次 **time / session weekday+period**

- Type: DayPeriod list
- Authoritative: **S2** (detail page) —347/347
- Fallback (labelled in output): none
- Conflict rule: Same as above: 0 of 732 in S1. X1 is an exact-match cross-check.

上课地点 / 教室 **room**

- Type: string, informational
- Authoritative: S2 for 地点
- Fallback (labelled in output): X1/X2 for 教室
- Conflict rule: Not part of any invariant. Both are reference-grade; label the origin in the UI.

限选 **capacity**

- Type: non-negative integer, or **unknown**
- Authoritative: **S2**

- Fallback (labelled in output): X1/X2 **cross-check only**
- Conflict rule: S2 wins. 0 / empty = **unknown** (never " unlimited" , never " full"); "/" = the official " no enrolment limit" —a different value from blank (A1 finding #3).

已选 **enrolled**

- Type: non-negative integer
- Authoritative: **S2**
- Fallback (labelled in output): X1/X2 cross-check only
- Conflict rule: S2 wins. Time-varying; always display with the capture time (§ 3).

教师 **teacher / chief**

- Type: string (public professional info)
- Authoritative: S2
- Fallback (labelled in output): X1
- Conflict rule: S2 wins. Instructor names are public course information; they are not used to build any per-person profile.

授课方式 / 考试方式 / 是否远程教学 / 助教 / 培养层次

- Type: enum/string
- Authoritative: **none on any official source**
- Fallback (labelled in output): **X2 reference only**
- Conflict rule: X2 is human-copied from a logged-in SEP view; A1's reverse-lookup found 0 hits for these words on all three official pages. Treat as unverified reference, never as a row-trustworthy fact; if it drives a decision, say so in the output.

培养方案学分下限 / 学位课规则

- Type: numbers + rules
- Authoritative: **S5** —sole source
- Fallback (labelled in output): none
- Conflict rule: No other source has it; where S5 is ambiguous it is a modelling gap (see degree-requirements.json known_modelling_gaps), not something to be filled from a course table.

” 我实际选中的课 / 最终结果”

- Type: personal
- Authoritative: **S3**
- Fallback (labelled in output): S2/S3 aggregate
- Conflict rule: Only S3 has it, and S3's raw files are never committed (§ 4). Post-2026-09-18 (system closed) it is only recoverable from an exported result.

报名窗口 / 纪律规则

- Type: dates + conduct
- Authoritative: **S6**

- Fallback (labelled in output): none
- Conflict rule: Quoted text; not re-derived.

冲突 Conflict

- Type: boolean
- Authoritative: **derived**
- Fallback (labelled in output): —
- Conflict rule: Never stored, never sourced: computed from two Slots by the three-condition rule (glossary §3, conflict-decision-table.md).

The one-line answer to " which source should I trust" : take S2 first; use S1 only to fill 开课学期 and to corroborate 属性/学分/学时; demote X1/X2 to cross-check and auxiliary; use S5 for the credit floors and S6 for the window and the rules; use S3 only for what actually happened to **my** plan.

3. Quality: known absences, sentinels, drift, expiry

3.1 Known absences (per field, per source)

Absence	Measured	Consequence
开课周 in S1	0 of 732	Nothing can be conflict-checked from S1 alone
星期节次 in S1	0 of 732	same
限选 / 已选 in S1	blank in the snapshot rows (e.g. "capacity": "", "enrolled": "")	no capacity pressure from S1
开课学期 in S2 / X1	no field	S1 is the only 学期 source; 6 of 14 planned courses have 学期 unknown from every source (plan-priority-summary.md)
campus in X1	no field at all	inferred from code position 18, always labelled
培养方案要求	absent from all of S1/S2/X1/X2	human input; frozen at data/s5-programme/degree-requirements.json
Join coverage	217 codes in both; 197 S1-only (no week/period/capacity at all); 130 X1-only (no campus, no semester)	for 197 courses a conflict check is impossible from the structured sources; that is a measured limit, committed as a number, not smoothed over

Absence	Measured	Consequence
S2 parse failures	12 recorded failures across terms (e.g. 180089050200MB001Y-201, 硕士学位英语 (慕课学习))	those courses exist but carry no parsed schedule; reported, not guessed
S2 spring 2026-27 (termId 89577)	0 rows , 158-byte file	2027 spring was unpublished at capture; any spring statement is out of scope

3.2 Sentinel values (this is where a literal reading produces a false headline)

- **capacity = 0 means "not set", not "admits nobody"**. 40 rows in the frozen snapshot have capacity 0 with a non-zero enrolment —one shows **86 enrolled** against it (180090125601M2004Y 财务与成本管理). invariants.md l3 records that reading this literally would have produced the wrong headline "the school oversells 40 sections"; after separating the sentinel: **real over-capacity violations (enrolled > capacity > 0) = 0**, and the honest conclusion is "cannot be determined from this field" —**not** "none exists". A1 separately observed genuine over-capacity (自然辩证法 422/420) from an earlier crawl, which is exactly why "0 violations" and "422/420" can both be true: different capture times.
- **Empty ≠ / ≠ 0**. Blank = unknown; "/" = the official "no enrolment limit"; 0 = not set. Three distinct meanings in one column; the checks report them in separate buckets so a sentinel can never be laundered into a violation count.
- **Stringly-typed numbers**. S2 gives "111", "22", "1.00"; S1 gives numbers. Compare numerically, keep the original string for the record.
- **Interval sets, not scalars**. 237 of 561 course records (42%) have discontinuous teaching weeks; this is normal, not an anomaly (A1 overturned claim #4).

3.3 Fields that move with time

capacity, enrolled, and therefore "remaining seats" and the whole **grab order**, are the only fields that are expected to change between two captures. Everything else (name, attribute, credit, room, code) is stable within a term.

- **Snapshot expiry risk**. The committed S2 snapshot was captured **2026-09-27**, which is **after** the registration window closed (2026-09-18 12:30). The conflict-check claim is unaffected (times do not move after publication), but the grab-order claim **cannot be validated against the committed snapshot at all**, and a later re-crawl would produce different numbers and different hashes.
- The model names this state `SnapshotStaled` (glossary §4): the plan is a hypothesis, not a fact. The only defence is that every output prints the capture time, and a stale one must be re-crawled.

- A1 measured the drift directly: 09-05 vs 09-15 enrolment figures differ, and the two plan versions differ by **22 removed / 45 added timetable cells**.

3.4 Format properties that are not guarantees

`invariants.md` §0 records that all 347 courses carry both a time and a week string, with matching ;-separated segment counts, and 0 unparseable segments. That is a property **of this snapshot**, not of the format: the check counts mismatches and would report them rather than assume them.

3.5 Provenance hygiene defects found while writing this document (recorded, not fixed here)

These are small but real, and they are the kind of thing a governance section exists to surface:

1. **Stale filenames in two .sha256 sidecars.** `data/derived/course-library-snapshot.json.sha256` names the file `yuquanlu-live-courses.json`, and `sep-aux-fields.json.sha256` names `aux-sep-fields.json`. The **hashes match the files they sit beside** (verified: 3724ade7..., 7d48eb75...), so integrity is intact —only the second column of the sidecar is stale from an earlier rename. A verification script that trusts the filename column would fail on a file that is perfectly fine.
2. **The 2026-27 captures (termId 89576/89577) are not covered by official-db-provenance.md**, whose six term sections stop at 84068 (2025-26 春). The 89576 file does carry its own `crawled_at` (2026-09-27 16:31:49) and a matching `.sha256`, so the record is not lost —but the provenance note is incomplete.
3. **`data/s5-programme/degree-requirements.json` had no .sha256 sibling** (closed on 2026-10-04: the sibling and `data/s5-programme/provenance.md` are now committed, and the source PDF's own sha256 is recorded there). The harder half of the gap stays open and is stated rather than papered over: the extraction was done **by hand** from a PDF that is not committed (a university document), so the hash pins the JSON and the identity of its source, not the fidelity of the transcription. This remains the one irreproducible link in the I2 credit-floor claim.
4. **X2 holds 2,951 course codes against 2,991 raw X1 entries** (the file is keyed by course code, so 40 duplicate codes collapsed). The file's own metadata does not state this; the count is a measured fact, not a documented one.
5. **S1 and S2 do not share a row schema.** S1 stores one coarse sessions [{room,time,weeks}] plus top-level time/weeks (empty); S2 stores a parsed schedule [{weekday,periods,weeks,rooms}] Any join must state which shape the field came from.

4. Retention

Three retention classes; **what is kept is a function of whether it can be re-derived**, not of whether it is interesting.

4.1 Long-term (kept indefinitely, with hash and capture time)

Frozen official/derived inputs and their `.sha256` siblings —the audit trail, and the reason any A1/A2 number can be re-checked years later. None of it is personal data; all of it is either published by the university or derived from published data.

Artefact	sha256	Why kept
data/s1-plan-workbook/ sections-snapshot.json (+ .sha256)	ffcf1ae0...	the planned-offering side of every join; 开课学期's only source
data/s2-public-listing/ *.json + history/*.json (+ .sha256 each)	e.g. 89576 6510aa2e..., 74468 a90296c9..., 84068 6e1f4505...	the current-state side; re-crawl would change hashes
data/derived/course- library-snapshot.json (+ .sha256)	3724ade7...	X1's frozen subset —the schema/sha are already cited, so it must not change
data/derived/ sep-aux-fields.json (+ .sha256)	7d48eb75...	X2 reference fields; frozen with a forced raw-hash check so the X1 chain stays intact
data/derived/ plan-priority-summary.md, credit-gap-summary.md, data/derived/a2/*	(regenerated by script)	aggregates only —counts, tiers, buckets; no row-level personal detail
data/derived/a2/ sample-plan.json	(regenerated)	an illustrative 6-course plan, explicitly not the real plan, so I1/I2 run from the repository alone
data/s5-programme/ degree-requirements.json, data/ s3-my-plan-versions/ plan-versions.md, doc/ evidence-ledger.md, doc/ data-authority.md	—	the rules and the claim→source→file mapping

4.2 Aggregates only (retained, but never at row level)

Anything touching a person: the plan's totals (14 courses, 26 course-learning credits, 25 evidenced-open in 秋季, 8 of 13 at $\geq 90\%$ capacity, P0/P1/P3/PX/PU = 3/3/1/1/6), the credit-gap verdict (2 categories short). These are kept because a claim needs a number; there is no per-course commitment anywhere in the repository.

4.3 Not retained (or local-only)

Item	Policy	Why
The two S3 .xlsx plan versions, the planner backup, the prototype HTML	local only, never committed (*.xlsx, 选课地图-本地备份 *.json, prototype/*.html, data/private/ are all gitignored)	personal data; the prototype embeds the plan
data/s2-public-listing/cache/ (1,400+ raw HTML pages)	delete after a successful freeze	re-fetchable and large; kept only so a crawl can resume
X1 raw download (8.1 MB)	not committed ; keep the hash 7c6090e4... only	re-fetchable; the hash is the evidence
The 4,877-record planner backup	not committed ; keep the hash 62594e79... only	contains the 开课校区 field for the whole school that S1's kept subset drops
data/private/ (credit-gap.md, plan-priority.md)	kept during the course, deleted at the end (§ 5)	per-course personal priority detail

4.4 Where it is retained

Public GitHub (github.com/Tom1246/ise-course) for § 4.1 and § 4.2; the author's local machine only for § 4.3. There is no third-party store, no cloud bucket, no analytics sink.

Retention is limited by history, and this must be said plainly: a file removed by a new commit still exists in the repository's history. § 4.1 / § 4.2 artefacts are chosen so that nothing personal has ever been committed; the gitignore in § 4.3 is the first line of defence, and the pre-commit protocol is not to stage anything under those patterns.

5. Deletion

A procedure for removing **one person's data** —written so it can be executed, not promised. It assumes the subject is the author (the only person whose data this project ever holds). Run

it in order; each step's check must pass before the next is attempted. **This document does not execute git operations**; steps 5 and 8 state the commands to run and their expected output.

Step 1 —delete the gitignored private working copies

```
rm -rf data/private/                # credit-gap.md, plan-priority.md (per-course)
rm -rf prototype/*.html prototype/*.png # the prototype embeds the plan; keep the plan
rm -rf data/s2-public-listing/cache/   # re-fetchable raw HTML, no personal content
```

Delete the personal working files outside the repository as well (the original imports and any exported copy). ~ globs are not recursive by default, so locate them explicitly rather than trusting **:

```
rm -f ~/Downloads/选课地图-本地备份*.json
rm -f ~/Downloads/选课规划-已选课程-*.xlsx
find ~/Desktop -maxdepth 6 -name '选课规划-已选课程-*.xlsx' -print -delete
```

Step 2 —confirm the personal freeze inputs are gone and only their hashes remain

The two S3 .xlsx files and the planner backup are the inputs to S1's import and to the plan-version report. They must be gone from every location, and the hashes recorded in data/s3-my-plan-versions/plan-versions.md (8d6f9e65..., dcc6cae7...) and doc/evidence-ledger.md (62594e79...) are the only thing that needed to survive:

```
find ~ -name '选课规划-已选课程-*.xlsx' -o -name '选课地图-本地备份 *.json' 2>/dev/null
# expect: no output
```

If a hash line is the only remaining trace, the deletion preserved the evidence and removed the data —which is the intended end state. If the files are still present, the greps in step 4 will pass over an empty repository while the data sits on disk, so this step is a precondition for step 4's result to mean anything.

Step 3 —confirm the repository now holds only aggregates

Read, do not assume:

```
python3 -c "import json;d=json.load(open('data/derived/a2/sample-plan.json'));print(d)"
# expect: the illustrative note + 6 codes that are explicitly NOT the real plan
```

Then open data/derived/credit-gap-summary.md and data/derived/plan-priority-summary.md and confirm every number is a count, a total, a tier or a bucket —no course name tied to the author, no per-course row.

Step 4 —git grep for the four string classes

Run each over tracked files only (git grep reads tracked files, so gitignored private files cannot create false confidence), from the repository root:

```
# The literal values are supplied locally and are deliberately not written down in the
# (writing them here would defeat the step that is meant to find them):
#     export SID='<student id>'  NAME='<name or pinyin>'  # then run the greps below

# (a) student ID
git grep -nE "$SID"

# (b) name / pinyin / handle
git grep -niE "$NAME|Tom1246"

# (c) programme code — ANCHORED, see the trap below
git grep -nE '(^|^[^0-9])0854[0-9]{2}([^\0-9]|$) '

# (d) local absolute paths / home-relative paths
git grep -nE '/Users/[A-Za-z0-9._-]+/|~/Desktop|~/Downloads|~/Documents|files/工作'
```

The trap in (c), measured: a naive 0854[0-9]{2} matches **658 lines** in this repository, because almost every course code embeds its programme code (180086085404P2003Y contains 085404). The anchored pattern above reduces that to **1 real hit**. An unanchored grep here is not a strict check —it is noise that trains the reader to ignore the output.

Measured result at the time of writing (A2, 2026-10-05) —non-zero, and that is the point of the step:

class	hits	where
(a) student ID	1	README.md:6
(b) name / handle	6	README.md:4,6,125 (byline + repo URL Tom1246), doc/assignments/a2/A2-summary-en-2026-10-04.md:31, doc/assignments/a1/A1-summary-en-2026-09-28.md:20, doc/archive/A1-brief-en-v1-2026-09-27.md:5 (Author: Liwei Tang)
(c) programme code, anchored	1	data/s5-programme/degree-requirements.json:9 (the programme string)

class	hits	where
(d) local paths	10	scripts/ {inspect_plan_versions, build_web_prototype, cred (the ~/Downloads/... defaults), doc/ selection-design.md:102, data/ s5-programme/ degree-requirements.json:4 (local_copy), data/s1-plan-workbook/ snapshot-provenance.md:3, and —before this pass —notebooks/ a2-domain-model-walkthrough.ipynb:36, which printed the machine's absolute home path into a committed notebook

A defect this scan actually caught: the A2 notebook's first cell once printed the machine's absolute home path into its stored output (a full `/Users/...` path, i.e. the author's local directory layout). The notebook now prints only the repository folder name, and the path is gone from the committed output. The finding is kept here because the scan is the reason it is gone —and because the same failure mode (a tool printing its working directory into a stored artefact) will recur in A3 and beyond.

These non-zero hits are the honest part of the procedure. As of A2 the repository still carries the author's name and student ID (README.md), the GitHub handle in four files, the programme code in the committed credit-rules file, and ten local-path strings —one of which is a full absolute path with the username. Most are deliberate (attribution, reproducible script defaults, provenance of an uncommitted source) and the student ID is the one that carries real risk. The deletion is finished only when every hit has been **classified** —keep with a stated reason, or remove and regenerate the artefact —not when the count reaches zero. A check that is written to expect zero, in a repository that legitimately contains attribution, will be disabled by the first person who sees it fail for a good reason.

Step 5 —re-scan committed PDFs page by page with pymupdf

The greps above see the `.md/.json/.py` text; PDFs hide theirs in a text layer. Extract and scan every committed PDF:

```
python3 - <<'PY'
import pathlib, re, fitz
pats = {
    "student_id": re.compile(os.environ.get("SID_PATTERN", r"\b\d{4}[A-Z]\d{10}\b")),
    "name":       re.compile(os.environ.get("NAME_PATTERN", r"$^"), re.I),
    "prog_code":  re.compile(r"(?:^[^0-9])0854\d{2}(?:^[^0-9]|$)"
```

```

    "local_path": re.compile(r"/Users/[\w.\-]+/|~/Desktop|~/Downloads|~/Documents"),
}
hits = 0
for p in pathlib.Path(".").rglob("*.pdf"):
    with fitz.open(p) as doc:
        for i, page in enumerate(doc, 1):
            text = page.get_text()
            for name, rx in pats.items():
                for m in rx.finditer(text):
                    hits += 1
                    print(f"{p} p{i} [{name}] ...{text[max(0,m.start()-40):m.end()+40]}")
print("TOTAL HITS:", hits)
PY

```

Run it **once per PDF set** —the A1 briefs are text-layer PDFs, so this is the step that would catch an ID or a personal course list that survived a Markdown edit. Each hit is then classified as (i) a public fact in an offer-set claim, or (ii) personal detail, which must be removed and the PDF regenerated.

Measured over the 9 committed PDFs (A2, 2026-10-05): 2 hits, 0 student-ID hits.

file	page	class	text
doc/selection-design.pdf	4	local_path	本地 (~/Desktop/ files/工作/硕士/)
doc/archive/ A1-brief-en-v1-2026-09-27.pdf	1	name/ handle	the first English draft's own byline

The student ID appears in **no** committed PDF —the same string that §5 step 4 finds in README.md. That is the point of scanning the PDFs separately: the two artefacts are generated from different sources, so passing one says nothing about the other.

Step 6 —scan the git history, not just the working tree

```
git log --all --oneline -- '*.xlsx' 'data/private/*' 'prototype/*.html' '选课地图-本地
```

If anything is returned, the data is in history even though the file is gone. Recovering it requires a history rewrite (`git filter-repo`) and a force-push —a decision for the author, and the reason the gitignore exists. Because this project's rule is "never commit it in the first place", the expected answer here is **empty**.

Step 7 —verify the tree is clean and re-run the checks

```
git status --porcelain          # expect: no modifications to tracked files (deletion
python3 scripts/a2_check_invariants.py  # still exits 0: the checks must be runnable
```

The point of step 7: deleting the personal data must not break the invariant checks. Because the checks run on `data/derived/a2/sample-plan.json` and the committed catalogue, not on the real plan, they still pass —which is itself the evidence that the repository never depended on the private data.

Step 8 —record the deletion

Append the date, the bytes removed, and the grep results to the A2 log so the deletion is verifiable after the fact. A deletion that is not recorded is indistinguishable from a deletion that was never run.

6. Privacy and access

6.1 Who may access what

Tier	Contents	Who can read it	What protects it
1 —public repository	S1/S2 frozen snapshots, S5 rules, derived aggregates, scripts, docs, the illustrative sample plan	anyone, including anonymous GitHub readers	nothing withheld that is personal; the repository is public by design
2 —gitignored, present in the local checkout	<code>data/private/</code> , the <code>.xlsx</code> plan versions, the planner backup, the prototype HTML/PNG, the raw crawl cache	only the author's local account	<code>.gitignore</code> patterns (<code>*.xlsx</code> , 选课地图-本地备份 <code>*.json</code> , <code>data/private/</code> , <code>prototype/*</code> , <code>data/s2-public-listing/cache/</code>)
3 —local-only, never inside the repository	the original imports in <code>~/Downloads</code> , the A2 working folder, the real plan	only the author	never staged; § 5 is the exit procedure

The unit of work is **one student, one semester, one plan** (glossary § 1) —no cohort, no department, so there is no product reason to hold anyone else's row.

6.2 What is in tier 1 that mentions a person

Instructor names (teacher, chief) in the S2 snapshot —**public professional information**, published by the university on a public page, retained verbatim as part of the official record, and used only to identify a course, never to build a profile. The author's own name/ID appear only in the README byline (see § 5 step 4, flagged).

6.3 Red lines (hard-coded, not policy aspirations)

1. **Do not collect another student's enrolment record.** Enrollment (student × section) is the identifying entity (glossary § 2); the crawl touches course-level public pages only and never a per-student view. This is A1's abuse scenario 2, and the answer is that the data is never read —not that it is read and hidden.
2. **Do not register on anyone's behalf.** The system has **no actuator**; it emits a timetable and an order only. Automatic seat-grabbing breaks first-come-first-served and is forbidden by the school notice (S6). The registration context is drawn as a dashed, out-of-scope box on the context map precisely so that adding a submit step is a visible scope change.
3. **Do not export an identifying roster.** Outputs are per-plan, for the plan's owner. No list of people, no student IDs, no names of students, no cross-student join. Where a person's data must appear in a committed artefact, it appears as an aggregate.
4. **No credentials, tokens or cookies** anywhere —no login is used by any committed script; the crawl reads public pages only.
5. **Do not let the AI fill a gap with a plausible value.** Every section must trace to a source + capture time; a mismatch is an error, not something to be smoothed over (A1's own AI misuse, and the reason `ai/ai-use-log.md` exists).

6.4 Access-control reality check

There is no authentication, no role model and no encryption in this project, and pretending otherwise would be worse than the truth. The whole access model is: **tier 1 is public, tiers 2-3 exist only on one machine, and the only thing that keeps them apart is the .gitignore plus the discipline of never staging a private file.** § 5 exists because discipline alone is not a control.

7. Threat to the claim

The A2 claim has two halves, and they are threatened by different fields. Stating which is which is how this document exposes rather than hides the weakness.

7.1 What breaks the conflict-detection claim

The claim: two sections clash iff same weekday $\cap$ overlapping periods $\cap$ overlapping weeks, and this is decidable from frozen committed data.

Field	If it is wrong or missing	Effect on the claim
开课周 weeks	a wrong or absent interval set	the claim fails in both directions: a bitemporal pair (自然辩证法 weeks 2-5/7-10 vs 中特 weeks 11-18 sharing cell C5) becomes a false conflict, and a genuine partial overlap is missed. This is exactly A1's observed failure —the AI read weeks wrong and produced a wrong recommendation. For the 197 S1-only codes the field is absent, so the claim is simply not decidable for them —a limit, not a bug.
星期节次 time	shifted by one period or one weekday	every pair in the catalogue changes membership; the 4,757 figure is a function of this field alone
课程编码 code	a wrong position-18 char, or a mis-join	silently moves a course in or out of the campus subset; two different sections could be merged into one identity
学分 credit 课程属性 attribute	wrong credit wrong value	only threatens I2 (the floor verdict), not I1 mis-buckets a course into " must take" vs " candidate" , which changes the plan under test

Mitigations already in the repository: the rule was validated on a 6-case decision table **before** the catalogue figure was quoted; the catalogue-level 4,757 is explicitly labelled offer-set density, **not** a violation of I1 (invariants.md I1 " Limit"); the plan-level check is the one that counts, and it is run on a committed illustrative plan so it is falsifiable from the repository alone.

7.2 What breaks the grab-order claim

The claim: the order to register in can be ranked from remaining seats plus semester-only availability.

Field	If it is wrong	Effect
已选 enrolled , 限选 capacity	stale by even one capture	the ranking is wrong; a full section looks open (A1's " the system cannot know by itself that the data is stale")
capacity = 0 sentinel	read as " full" or as " unlimited"	the order is wrong in two opposite ways
开课学期 semester	unknown (6 of 14 planned courses)	a spring-only course is ranked as if it could be deferred, or an autumn-only one is not prioritised

This is the weakest half of the deliverable, and the repository says so on its face: the committed snapshot is captured **after** the registration window closed, so the grab-order claim **cannot be validated at all** against the committed evidence. The only honest status of that claim is " the inputs and the ranking script are committed; the ranking was not re-checked against a live system" . `model-walkthrough.md` " what the walkthrough did not settle" says the same thing: the ordering rule is deliberately left to A3/A4 rather than specified now.

7.3 How this document exposes rather than hides the weakness

- **Unknown is a first-class value.** Missing capacity is reported as " unknown" , never as unlimited, never as full; the sentinel bucket is printed **separately** from the violation bucket so it cannot be laundered into a headline.
- **Every finding carries source + capture time + hash.** A stale conclusion is visible as a stale timestamp, not as a wrong number.
- **Limits are written next to the claims.** `invariants.md` I1 (4757 is not a violation), I2 (floors only —the ceiling is left unimplemented rather than invented), I3 (" cannot be determined" ≠ " none exists").
- **Negative results stay in.** The coverage gap (197/130), the 12 parse failures, the expired snapshot, the overturned AI claims —all committed.
- **Provenance hygiene defects are recorded** (§3.5), including the stale `.sha256` filenames and the missing provenance note for the 2026-27 captures.
- **Deletion is executable** (§5) rather than promised, and §5 step 4 states its own expected non-zero hits instead of pretending the repository is clean.

7.4 The single weakest conclusion in this document (stated plainly)

The weakest statement here is " **0 real over-capacity violations**" in §3.2. It is true of the frozen snapshot and it is arithmetically checked, but it is the most fragile claim in the document: the same school produced **422/420** in an earlier crawl (A1), the sentinel bucket holds 40 rows whose capacity is unknown, and the snapshot is post-window so it cannot be refreshed to settle the question. " 0 violations" is therefore a statement about **one capture of a field that is frequently left blank**, not a statement about the university's behaviour.

Anyone reading it as " the school does not oversell sections" would be reading a governance limit as a fact —which is precisely the error this section exists to prevent.

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A2 —Model walkthrough record

A walkthrough is done by taking the A1 scenarios and running the A2 model against them, in order, and writing down every place where the model was wrong, ambiguous, or silent. The findings below are the record; several of them changed the model, and those changes are noted in place.

- Model under test: `domain-glossary.md` (language), `figs/*.dot` (structure), `data-dictionary.md` (field reality), `invariants.md` + `conflict-decision-table.md` (rules).
- Data under test: the committed frozen inputs (`data/derived/course-library-snapshot.json`, `data/s1-plan-workbook/sections-snapshot.json`, `data/s5-programme/degree-requirements.json`) —no live system.
- Executed with: `python3 scripts/a2_check_invariants.py`, plus the SQL-free queries recorded inline.

Scenario 1 —normal: plan one semester from the published documents

Walk: `CoursePlanPublished` → read the workbook (732 Yuquanlu records, 413 course codes) → `CatalogueSnapshotTaken` from the listing (347 courses with slot and capacity) → parse slots → apply the decision table → emit a timetable and a grab order.

Findings:

1. **The two upstream sources do not agree on what exists.** A1 measured this as 282 listing codes vs 234 workbook codes with 223 in common. The model must therefore treat " the catalogue" as the union of the sources with an authority per field, not as a single table —which is what `data-governance.md` § 2 now specifies. **Model change:** the context map shows two upstreams, not one.
2. **" The course clashes" is almost never true.** A course normally has several sections (`-01`, `-08`, `-W003`), and the clash is a property of sections. An early draft of the glossary said " 课程冲突" ; it is wrong and is now listed among the words the model refuses to use loosely.
3. **The workbook cannot answer " when" .** 0 of 732 records carry weekday/period or week ranges —the fields exist but are empty, and the empty sessions array is the evidence. A model that let the workbook be authoritative for slots would be silent exactly where the student needs an answer. **Model change:** field-level authority (`data-governance.md` § 2) instead of source-level.

Scenario 2 —boundary: half-semester courses, a full section, a stale snapshot

Walk: rows 3 and 4 of the decision table; SectionFull; SnapshotStaled.

Findings:

4. **Row 3 is not an edge case, it is the norm.** Slots are routinely split into week blocks (第 2–5, 7–17 周). A screen that ignores weeks marks most of the catalogue as mutually exclusive. This is why the decision table, not a rule of thumb, is the specification.
5. **Two courses may even share one slot with different week blocks** —the data contains 周一 (10–12); 周一 (10–12) with two different week ranges on the same weekday and periods. The model therefore stores a set of slots per section, and the UI must show the week block with every slot, not just the day.
6. **capacity = 0 is a sentinel, not a capacity.** The first run of the I3 check reported "40 violations" —all of them rows with capacity = 0 and real enrolments (up to 86). The model now states capacity as integer or unknown, and the check reports sentinel rows and real over-capacity rows separately. **Model change:** I3 was rewritten; see invariants.md.
7. **A stale snapshot is a first-class state, not an accident.** The listing is a snapshot taken 2026-09-27 and registration continues until 09-18 (···) —capacity keeps moving after the snapshot. SnapshotStaled is a domain event, and the plan it produces is a hypothesis. A1 observed the same section at 422/420 in one capture and differently later, which is why every frozen input carries a capture time and a hash.

Scenario 3 —failure: the real plan that had to be rearranged

Walk: SectionFull → SectionSwitched → PlanRevised (the A1 record: 自然辩证法 moved from 周二第 1 节 + 周日第 1 节 to 周三第 5 节 + 周六第 5 节; 中特 from 周二第 1 节 to 周三第 1 节).

Findings:

8. **A switch is a multi-section event.** Moving one course changed three others; the model must re-run the whole plan check after every edit, which the temporal property in conflict-decision-table.md §3 states.
9. **The clash the plan actually contained is reproducible from committed data:** 计算机代数 / 数字图像处理与 分析 / 深度学习 all at 周四第 10-12 节 → 3 conflicting pairs. The walkthrough closes the loop between the A1 narrative and the A2 model: the narrative's claim is now a check, not a story.

Scenario 4 —misuse: " just register for me"

Walk: the boundary where the model refuses. See A1's abuse scenario: an automatic seat-grabbing bot breaks first-come-first-served and the school notice explicitly forbids disrupting or hoarding registrations.

Findings:

10. **The model has no actuator and must not grow one.** The deliverables are a timetable and an order; the registration context appears on the context map as a dashed, out-of-scope box precisely so that ” add the posting step” is a visible scope change, not an implementation detail.
11. **The privacy boundary is part of the model, not a policy afterthought.** Enrollment is the identifying entity, so per-course and per-student detail lives outside the repository (data/private/, gitignored) while only aggregates are committed; the deletion procedure is written down in data-governance.md §5 so it can be executed rather than promised.

What the walkthrough did not settle

- **The grab order is under-specified by this model.** ” Which section first” depends on capacity pressure that is itself stale; A2 stores the inputs (capacity, enrolled, timing) but does not yet define the ordering rule. That is deliberate: it belongs to A3/A4, and pretending to specify it now would be a specification that cannot be tested.
- **SectionOverCapacity has no decision attached.** The model flags it; whether the planner should avoid such a section is a product decision this walkthrough did not make.
- **Spring-term data is thin** —2027 spring is unpublished, so the model's ” semester” value is exactly the case the data dictionary labels unknown, not a value to be guessed.

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Appendix —diagrams (source and render)

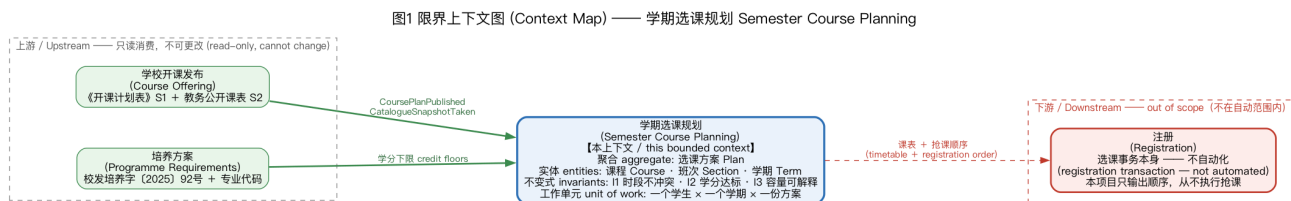


Figure A1 —Bounded contexts: two upstream sources, the planning context, and the registration context that stays out of scope.

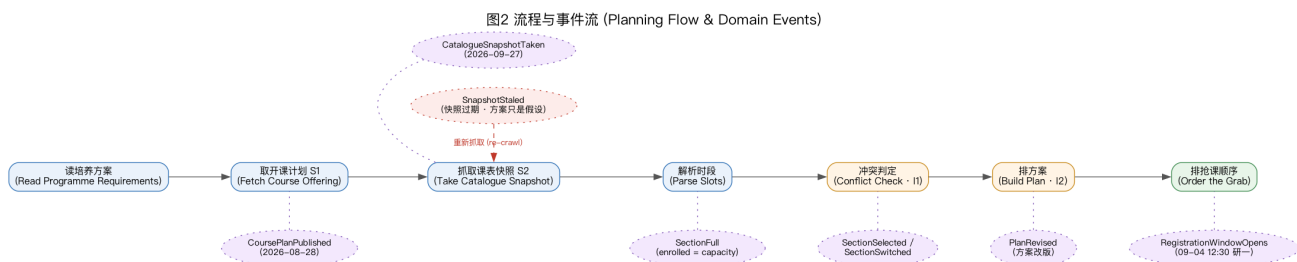


Figure A2 —Planning flow with the domain events attached to each step.

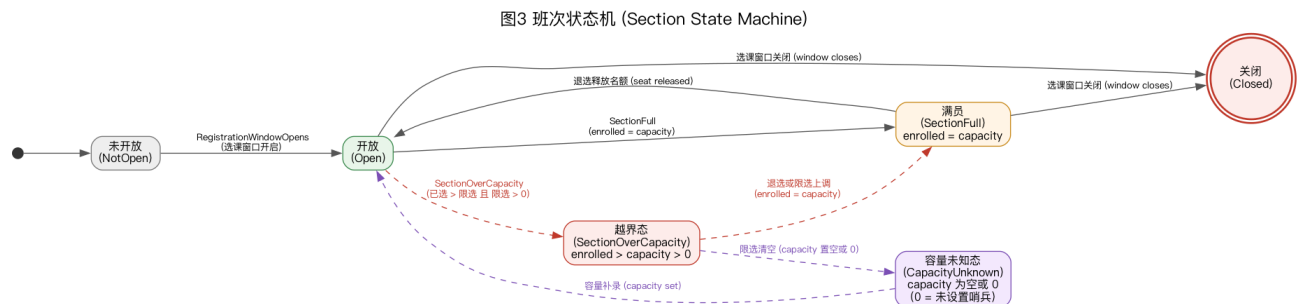


Figure A3 —Section states, including the over-capacity and unknown-capacity states.

图4 选课记录状态机 (Enrollment State Machine)

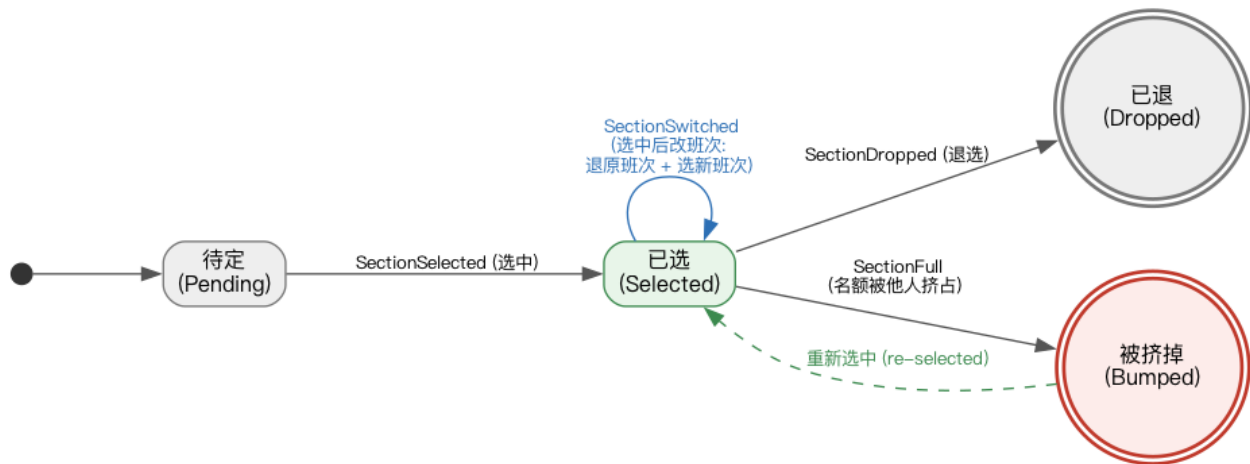


Figure A4 —Enrollment states, including the switch that re-arranges a plan.

Each figure is generated from the committed `.dot` source by `bash figs/render.sh`; the sources are the authoritative artifact and the PNG is a build output.